Dynamics of Financial Aid Tournaments

Arkādijs Zvaigzne
Harvard University

November, 2025

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Abstract

Financial aid programs in higher education vary widely in design, including how aid is structured and the timing of provision. This paper studies the impact of financial aid provided as a repeated tournament and its dynamic treatment effects. I exploit a relative GPA-based eligibility rule in a regression discontinuity design to estimate the causal impacts of two types of aid - a tuition waiver and a stipend on top. Waivers have powerful effects on the extensive margin, increasing graduation rates by 12pp, and GPA by 0.4σ. Stipends affect the intensive margin by increasing student GPA in the next semester by 0.3σ and the extensive margin by increasing graduation rates by 8.8pp. I find a powerful crowding-in effect, where receiving aid in one semester significantly increases the probability of receiving it in the future. Decomposing the impact reveals that a substantial portion of the total long-term benefit of aid comes from the crowding-in of future resources, suggesting that static analyses may underestimate the full value of financial aid programs.

JEL Classification: H52, H4, I21, I22, I23, I28

Keywords: Student Financial Aid, Higher Education

*I am grateful to Riga Stradiņš Univeristy and the Univeristy of Latvia and especially Ilze Astašova, Sanita Liepiņa, and Ineta Romanovska for their invaluable assistance in acquiring the data as well as consultations on the institutional context and nuances. I received helpful comments from Peter Blair, Susan Dynarski, Eric Taylor, Richard Murnane, Luke Miratrix, Alexander Bolves, Julian Hayes. I also thank the participants of the BE-lab, conference participants at AEFP 2023, and participants at the Stockholm School of Economics in Riga Open Workshop for feedback on this paper.

†Doctoral Candidate at Harvard Graduate School of Education. Email: azvaigzne@g.harvard.edu

1 Motivation

Governments worldwide invest substantial public resources in higher education, viewing it as a primary engine for producing the human capital that drives modern economic growth (Deming, 2022). Many talented individuals are credit-constrained and cannot finance the optimal level of education on their own, and the social returns to education often exceed the private returns, leading to underinvestment without public support.

To address these issues, countries spend vast sums on student aid; in the United States alone, this figure reached $256 billion in the 2023-2024 academic year (Ma et al., 2024). Evaluation of financial aid programs has found that these investments work - on average, an additional $1'000 in aid increases student GPA by up to 0.16 standard deviations, persistence by up to 5 percentage points,and graduation by 5 percentage points (Dynarski et al., 2023; Nguyen et al., 2019).

Despite this massive scale of investment, our understanding of aid's long-term effectiveness is clouded by the complexity of real-world policy environments. In most developed economies, students navigate a fragmented landscape of interacting federal, state, and institutional aid programs. This fragmentation creates a central, unresolved puzzle regarding the net dynamic effects of financial aid: Do these overlapping programs create a crowd-in where one award facilitates future support, or do they simply crowd out one another, where one award displaces another and yields little net benefit to the student?

This paper overcomes these challenges by studying a unique, tournament-style financial aid system in Latvia where competitively reassigned aid is the sole source of support more than 90% students. By exploiting a rank-based eligibility rule in a dynamic regression discontinuity design, I causally identify the long-term effects of aid and, crucially, decompose them into a direct marginal impact and a crowd-in effect.

My analysis is guided by a conceptual framework that synthesizes four key elements. First, policymakers are interested in both ensuring that students increase their performance during their studies and accumulate more human capital (the intensive margin), and that they persist in their studies (the extensive margin). This maps directly onto a central design challenge in personnel economics: how to structure a tournament to both elicit maximum effort and ensure continued participation, preventing competitors from dropping out. Second, students are credit-constrained, which leads them to underinvest in their own human capital (Becker, 1964; Lochner & Monge-Naranjo, 2012; Rosen, 1976). Third, financial aid is allocated via a rank-order tournament, an intervention that both alleviates these binding constraints for winners and creates powerful performance incentives for all competitors (Lazear & Rosen, 1981; Rosen, 1986). Fourth, the repeated nature of this tournament creates a dynamic feedback loop.

I posit a crowd-in mechanism where winning aid in one period improves a student's performance, thereby endogenously increasing their probability of winning aid in subsequent periods. This virtuous cycle, where aid begets more aid, is a specific form of dynamic complementarity, in which an initial investment increases the productivity of all future efforts (Cunha & Heckman, 2007). This paper tests this hypothesis directly, examining whether the impact of aid is magnified for students with higher existing levels of human capital. I find evidence for this mechanism which implies that aid does not simply reward high-achievers but actively multiplies their success, linking the tournament's incentive structure directly to the technology of skill formation. The phenomenon of tournaments multiplying success has also been shown to exist in labor market promotion tournaments (Merton, 1968; Rosen, 1986).

Answering this question is empirically challenging. Studies in multi-program systems like the U.S. can at best estimate a net effect that is an unknown combination of positive complementarities and crowd-out wherein financial aid programs account for dollars received from other sources and decrease their contributions (Eng & Matsudaira, 2021). Furthermore, most prior work has focused on estimating the static effects of an aid award or the impact of simple renewal requirements (Scott-Clayton, 2011), and has been unable to causally decompose the total long-term impact of aid into its distinct dynamic components.

I use rich, administrative student-level panel data from two of the three largest universities in Latvia for a total of 198,286 student-semester observations. In 2022, the two institutions enrolled 326 of all tertiary students in Latvia. The data contain semester-by-semester information on students' grades, aid status, program of study, and graduation outcomes.

A critical feature of this institutional setting is that for over 90% of students, these government-provided waivers and stipends are the only form of financial aid available, which allows me to study the aid's effects in isolation. In the US, by contrast, aid is awarded by multiple levels of government, foundations, and by schools themselves. In the US aid process, an increase in one source of aid often crowds out another source - and this is typically not observed by the economist. This would tend to bias downward the estimated effect of a dollar of aid (Eng & Matsudaira, 2021).

The aid amount is the same each semester and is competitively reassigned based on a student's relative GPA rank, creating a stable, repeated tournament. This higher education funding model is not unique to Latvia, but is representative of many post-Soviet nations, such as Ukraine, Russia, and Kazakhstan, enhancing the external validity of the findings (Smolentseva et al., 2018).

My identification strategy exploits the institutional rule that allocates aid based on this GPA rank within a cohort in a regression discontinuity (RD) design. As the probability of aid receipt changes discontinuously at the cutoff, I use this variation caused by the rank distance from the cutoff as an exogenous instrument for aid receipt. Furthermore, students within these programs follow a fixed curriculum, which limits their ability to strategically alter course difficulty in response to aid incentives.

My key methodological contribution is the use of a dynamic RD decomposition method (Biasi et al., 2024; Cellini et al., 2010; Taylor, 2014) to go beyond estimating a single treatment effect. This approach allows me to recursively estimate two key parameters: the total long-term effect of an initial aid award, and the crowding-in parameter, which measures the effect of winning aid today on the probability of winning it in the future. This decomposition lets me isolate the marginal effect of a single aid award from the value generated by the future aid it helps a student secure.

I find that competitively reassigned aid has large and significant total effects on student success. Receiving a tuition waiver has impacts on both the extensive and intensive margin, increasing student GPA by 0.43σ and increasing graduation by 11.9 percentage points (pp). Stipend receipt also further impacts GPA, increasing it by 0.2σ and graduation rates by 8.8pp. These effects are at the upper end of those found in the prior literature (Dynarski et al., 2023; Nguyen et al., 2019), highlighting the power of aid in a setting with strong incentives and limited outside options.

My central contribution is the identification of a powerful and persistent positive crowd-in effect. Winning aid in one semester significantly increases the probability of receiving it in subsequent semesters, with effects persisting for up to three years. My dynamic decomposition reveals that this crowding-in of future resources is a substantial driver of the large long-term benefits of aid, particularly for awards granted early in a student's academic career.

This paper makes two primary contributions. First, to the literature on student financial aid (Dynarski et al., 2023; Nguyen et al., 2019), I provide the first causal estimates of a crowd-in mechanism. By measuring these dynamics in a clean policy environment, my findings offer a crucial benchmark for the full potential of aid and suggest that static analyses in more complex settings may underestimate the total value of these programs by overlooking powerful, positive feedback loops.

Second, I contribute to the literature on personnel economics and tournament theory (Bull et al., 1987; Drago & Garvey, 1998; Lazear & Rosen, 1981). My paper moves beyond the traditional empirical settings of executive compensation, sales contests, or professional sports, which often face challenges with subjective performance metrics or limited generalizability. The educational competition studied here serves as a unique natural laboratory with objectively measured performance (GPA), explicit rules, and high stakes, allowing for a cleaner and more robust test of the theory's predictions.

Specifically, this study provides rare causal evidence on the mechanics of dynamic tournaments. The analysis of the crowding-in effect, where success begets future success, empirically measures the option value of winning an early stage of a multi-period competition - a core concept in the theoretical work of Rosen (1986). Furthermore, a central challenge in tournament design is maintaining participation and mitigating the discouragement effect that can lead less-skilled competitors to withdraw. By directly measuring student persistence as the participation margin, this paper offers new insight into how prize structure and contest design can affect individuals engagement in the competition over the long run. Finally, by integrating the concept of dynamic complementarity, the paper shows how tournament incentives interact with human capital accumulation, demonstrating that the returns to winning are amplified for higher-skilled participants. These contributions offer a uniquely detailed and dynamic view of how tournament incentives shape behavior and long-term outcomes.

2 Institutional Background and Data

2.1 Student Financial Aid in Latvia

The higher education system in Latvia is concentrated, with the three largest public universities enrolling half of the 76'000 total students in the country. The funding system is organized based on the dual-track tuition model, wherein some students receive tuition waivers and all other students pay full tuition. This model is common throughout all but one of the post-Soviet countries, as well as in some postsocialist countries in Central and Eastern Europe (Smolentseva, 2020).

In Latvia, the primary forms of student financial aid are merit-based government stipends and tuition waivers for students.[1] Tuition waivers are provided to 41% of all students, and 13% of all students received stipends in 2017. In the same year, only 2.8% received financial aid from any other source. Nearly half of the students get no financial aid.

The tuition waivers are allocated to specific programs and cohorts, creating financial aid tournaments for tuition waivers and stipends within cohorts. The government-provided financial aid for a large group of students is regularly reassigned - tuition waivers are assigned for the rest of the academic year, and stipends are assigned for one semester. All students who study in full-time programs for which the government provides waivers have a chance of acquiring a tuition waiver for the next semester. Students who receive tuition waivers also have the opportunity to compete for stipends. These stipends are paid out to students as direct money transfers to their bank accounts every month for the duration of the stipend.

The government requires the aid it provides to be allocated based on merit. Merit for incoming students is defined as their performance in high-school exit exams. The students with the best grades in the exams are rank-ordered within their admitted program and receive tuition waivers. In subsequent semesters, institutions have freedom to define merit as GPA with with an optional addition of other characteristics such as research output.

Tuition waivers and stipends are also extensively available, with roughly half of the bachelor's and master's students in public universities receiving tuition waivers and 9% of undergraduate students and 7% of graduate students receiving stipends. The average student in 2022 had a total monthly expenditure of €1'356, of which tuition fees for the average full-time fee-paying student comprised €313 (Koroļeva et al., 2022).[2]

The government also provides stipends that are meant to be disbursed to the highest-

[1] Within the OECD, about half of the countries offer scholarships awarded on the basis of primarily merit (OECD, 2022).

[2] The other top monthly expenditures were board(€372), room (€365), and transportation (€106).

achieving students receiving tuition waivers. These stipends pay EUR 99.60 per month for the five-month duration of the semester. This stipend gets deposited into the student's bank account. The stipends cover roughly 10% of the average student's monthly expenses, excluding tuition.

Students within Latvian higher education typically follow a predetermined sequence of courses each semester, a common characteristic of many European higher education systems. This fixed curriculum means that academic competition for aid occurs over a common set of academic requirements for all students in a cohort.

The student financial aid assignment mechanism is visualized in figure 1.

[Figure 1 about here.]

The centrally planned nature of the Soviet Union and the centralization that came from it meant that all the republics within the Soviet Union had broadly similar higher education systems, characterized by having free higher education and being linked to the needs of the centrally planned economy. After the collapse of the Soviet Union, the 15 now independent countries have retained similar institutional arrangements to this day (Smolentseva et al., 2018). From Latvia to Ukraine to Turkmenistan, 11 of the 15 countries of the former USSR provide state-funded seats that waive tuition to students, as well as stipends that are provided to students based primarily on merit (Smolentseva, 2020). Thus, the Latvian institutional context is representative of other countries of the former USSR.

2.2 Difference Between the Two Institutions

2.2.1 Riga Stradiņš University

Riga Stradiņš University (RSU), is the third-largest institution in the country, enrolling a total of 8,766 students in 2018. The institution itself is under the oversight of the Ministry of Health and specializes in healthcare studies. A unique feature of the reassignment mechanism in RSU is that it guarantees waivers to students if they acquire them upon enrolment, with students not having to compete and also being guaranteed a waiver if they return to studies after stopping out. The waivers that are reallocated are thus from students who received a waiver upon enrolment, but then dropped out or stopped out in some later semester.

To illustrate this mechanism, a student who applies for an undergraduate program in Pharmacology would then compete with his potential cohort mates for the available tuition waivers. The first assignment would occur based on student high-school exit exam scores.[3] However, should a tuition-waiver-receiving student decide to drop out or stop out, their tuition waiver would then be up for reassignment. The reallocation would only occur within the same program and cohort. In an average year, around 40% of students are studying in waivers that they acquired upon enrolment, and 2% study in reassigned waivers.

RSU uses additional criteria for stipends. Stipends are allocated based on the GPA of the previous semester of students and their scientific output such as conference presentations, participation in research groups, publications, and patents. Generally, only medical students actively participate in research and do so after their third year of study.

2.2.2 The University of Latvia

The University of Latvia is the the largest university in the country that enrolled 20% of all students in Latvia in 2024. It is a classical university, offering a broad range of programs from IT to philology to physics to medicine. It is under the oversight of the Ministry of Education and Science. Compared to RSU, which is primarily healthcare-focused, UL has a broader range of fields of study and a larger distribution of students across them, with a smaller average tuition fee due to healthcare programs being more expensive.

The University of Latvia also has slightly lower achieving students, with the average

[3] The first assignment for graduate students occurs based on their average GPA in their undergraduate studies, as well as based on admission exam scores if there are admission exams for that program

mathematics high school exit exam score of its students being 49% (roughly corresponding to the 54th percentile of all those who enroll in higher education), whereas for RSU it is 53% (59th percentile). Furthermore, acquiring a tuition waiver at RSU is more difficult, as the average tuition-waiver-receiving student at RSU is at the 63rd percentile, and the average at UL is at the 59th for math.

The stipend receipt policy in UL is similar to RSU, wherein state-provided tuition waivers and stipends are allocated to students based on their academic performance. The key difference is that UL uses only student GPA in their financial aid provision, creating a sharp cutoff.

2.3 Data

To estimate the effect of acquiring and losing various forms of financial aid for students, I use data provided by RSU on students in all bachelor's and master's healthcare programs from 2015 to 2022, a total of 33,269 unique students that are observed each semester. This data includes semester-by-semester information on the grades, program, semester, year, graduation, tuition waiver, and stipend status of the students, as well as limited demographic characteristics such as gender, age, and school attended.

For UL, I employ data from 2010 to the second half of 2020. I use data for all tuition-waiver-receiving students for a total of 63 925 student-semester observations. I use this to both explore the external validity of my findings, as well as to increase the statistical power of my estimates.

One challenge with the UL data is that it is only available for students who are eligible for stipends (meaning those who received waivers), and I do not observe students who are on both sides of the waiver reception margin. Furthermore, I do not observe student demographic data, which limits opportunities for a balance test.

Table 1 provides the descriptive statistics for RSU and 2 provide the descriptive statistics for UL.

[Table 1 about here.]

[Table 2 about here.]

The institutions do not store data on whether a student applied for a tuition waiver or stipend and their final evaluation in points after combining the GPA with other criteria. This creates measurement error, as I only observe one of the components of the full running variable that is used to assign aid. The potential rank of each student is calculated based on only their GPA and whether they are eligible to apply for the scholarship or not.

The institutions also do not store data on a student's admission score. However, I observe whether a student received a waiver upon enrollment and student GPA, permitting me to evaluate the impact of aid in subsequent semesters after enrollment.

The students compete within their cohorts in a given semester - the risk set. As the programs are standardized with students taking a pre-determined list of courses in predetermined semesters, if a student does not complete their semester, they will have to retake their semester and thus be moved to a lower cohort. The risk set is dynamic and changes every semester for a student as they might fall behind, stop out, or they might have students who fell behind or stopped out come into their risk set.

I group individuals by risk sets and then take the subset of individuals who are eligible for each type of financial aid. For stipend risk sets, these are individuals who have received tuition waivers, as it is a precondition for eligibility. For tuition waiver risk sets, these are individuals who did not receive tuition waivers upon enrollment.

As the stipends are assigned based on rank order, I create a running variable that is based on rank and recenter it around the last person to have received financial aid so that the individual who received aid with the lowest rank in their risk set is at 0.

The summary stats of the full data and of the marginal stipend and waiver competitions are reported in table 1.

For RSU, using student secondary school data, I further create two demographic variables - whether a student attended a school that provides instruction in a minority language, and whether the school is located in the capital. In Latvia, Latvian language proficiency is a significant requirement for reception of aid (waivers and stipends are only provided in Latvian-language programs), and plays an extremely important role in the job market. Regional disparities are very high in Latvia, with the capital being considerably richer and having a higher level of GDP per capita than other regions. These two variables help to approximate students' social and economic statuses.

3 Estimating the Impact of Financial Aid

Separating out the direct treatment effects and dynamic treatment effects requires using both a regression discontinuity design and a dynamic regression discontinuity design. The regression discontinuity design provides me with estimates of the impact of aid that include the impact of aid crowding in future aid. The dynamic regression discontinuity design allows me to separate the impact into the marginal effect and the crowding-in effect of aid.

3.1 Estimating the treatment effects

The identification strategy exploits the institutional rule whereby eligibility for tuition waivers and stipends is determined by a student's rank based on academic performance (GPA) relative to a cutoff within a defined program-cohort-semester risk set. This rule generates discontinuities in the probability of aid eligibility at the cutoff rank, providing quasi-experimental variation that can be used by an RD design. The parameters identified represent Local Average Treatment Effects (LATE) specific to students near the eligibility cutoff whose treatment status is determined by crossing this threshold.

To recover the estimates of the effect of various types of aid on student outcomes, I use a regression discontinuity design that compares students just below and just above the grade cutoff. The identifying assumption of my estimation strategy is that potential outcomes are continuous throughout the cutoff, i.e. that they do not discontinuously "jump" (Lee & Lemieux, 2010). This means that students near the rank cutoff are comparable in their potential outcomes regardless of reception of treatment, allowing me to estimate the effect of the provision of a tuition waiver or stipend on their academic outcomes.

3.1.1 Estimating Static Impacts of Aid

I first begin by estimating the static impacts of of aid receipt on student outcomes. Let r_{ijt} denote the rank of student i in risk set j during semester t, and let $Above_{ijt}$ be an indicator variable equal to 1 if r_{ijt} meets or exceeds the cutoff rank c_{jt} (i.e., $r_{ijt} \geq c_{jt}$, assuming higher rank is better) and 0 otherwise. The rank r_{ijt} serves as the running variable. The central identifying assumption is the continuity of potential outcomes at the cutoff c_{jt}, conditional on the running variable (Lee & Lemieux, 2010). Under this assumption, students near the cutoff are comparable, differing primarily in their aid eligibility status induced by the discontinuity.

I use rank as the running variable as that is the one that most closely mirrors the assignment mechanism. I empirically validate this, with rank as the running variable having the strongest predictive power. The main challenge with using rank as the running variable is that it is explicitly discrete. However, using GPA has the same challenges.

I employ local linear RD methods to estimate two fundamental parameters of the dynamic system. The total effect ($\delta^T_{t,t+k}$) represents the average causal effect for individuals near the cutoff of crossing the eligibility threshold in semester t on an outcome Y measured k semesters later ($Y_{i,t+k}$). This parameter captures the full impact of initial eligibility, including any effects mediated through induced changes in subsequent aid eligibility, and is estimated by the following equation:

$$Y_{i,t+k} = \alpha + \delta^T_{t,t+k} Above_{ijt} + f(r_{ijt}) + \theta_{jt} + \epsilon_{ijt} \tag{1}$$

where $f(r_{ijt})$ is a linear function of rank, and θ_{jt} represents risk set fixed effects.

I use data-driven bandwidth selection (Calonico et al., 2020) and robust standard errors clustered at the student level. I then perform standard RD validity tests, including density continuity tests (McCrary, 2008) and covariate balance tests to assess the plausibility of the identifying assumption.

For RSU, given the less sharp cutoff, this approach does not create a perfect threshold, as, as individual applications and their scores by the committee deciding on aid are not observed. At the same time, for UL, given that aid is provided only based on GPA, the cutoff is quite sharp.

There are several threats to validity to this approach. The first threat is that students could for some reason have different potential outcomes across the threshold. Although I cannot empirically test this assumption, I can build intuition on it by testing whether the observed characteristics of students are continuous throughout the cutoff. Table 3 reports the result of the balance test. What I find is that students' demographic characteristics do not vary across either threshold, which lends credence to the assumption that the group just above and just below the cutoff might be comparable. Due to a lack of demographic data, I am not able to conduct a balance test for UL. [4]

[Table 3 about here.]

The second threat is that students could potentially self-select into receiving aid. This would lead to individuals above the cutoff being qualitatively different from those below it, meaning that they would not be comparable. In my context, this would require two things - that students choose whether to apply or not apply for aid and that they can precisely manipulate or tell what their grades and rank will be in their group.

Although students can choose to apply or not for a scholarship, this condition is necessary but not sufficient to invalidate this approach. To be able to perfectly self-select, the

[4] I have negotiated access to additional demographic data from UL and hope to receive it and conduct a balance test for it in the next version of the paper.

students would need to have perfect information on the GPA of their cohort mates and whether their cohort mates will apply for aid or not. The institution does not provide information for students on the grades of their peers and their rank in their group.

An alternative risk is that administrators could potentially selectively encourage students to apply for financial aid. For example, administrators might discriminate against individuals based on their characteristics and specifically encourage certain groups to apply. However, to do so, the administrators would have to know the student's full GPA and the GPA of their peers and be able to perfectly manipulate students near the threshold. Given the centralized nature of financial aid allocation and that students submit them electronically to the central administration and not within their departments, it is highly unlikely that administrators would be able to clearly manipulate the marginal student, as they would now know who the marginal student is. Evidence in support of this is also given by the balance test, as, if administrators selectively encouraged students to apply, I would expect discontinuities in student characteristics across the threshold.

One way to test for selection into treatment is through conducting a density test, wherein the distribution of grades just below and above the cutoff is compared (Cattaneo et al., 2020; McCrary, 2008). Although this test is useful for continuous data, the way that the GPA is constructed in my setting does not lend itself well to the density test. The aid is assigned using semester GPA, which is the average of the GPA of courses taken that semester. The policy is that the grade for a course can only be a whole number, which creates sharp discontinuities in the densities as shown in figure 2.

[Figure 2 about here.]

Due to the structure of the data, a regular density test would be uninformative, as it would be picking up the discontinuities in the densities due to how the data is generated. At the same time, it is also possible that there are discontinuities in the densities and that students are also selecting into treatment. To estimate the probability of this occuring purely due to choppiness of the data, I use a permutation test.

To estimate the probability of the density test failing due to the structure of the data, I use a permutation test. As the cutoff is set for each separate risk set at a specific value, I resample all of the cutoffs for the risk sets and reassign them to other risk sets. If students are selecting into treatment then the density test on the observed data should be a very extreme value. On the other hand, if the changes in densities are due to the structure of the data, then the observed values would not be extreme. I report the results of these tests in figure 3 for RSU and in 4 for UL.

[Figure 3 about here.]

[Figure 4 about here.]

Using the permutation approach, I fail to reject the null hypothesis that the observed value of the density test is different from a dataset where the cutoff is as good as random. Additionally, in section A in the appendix, I show that the distribution of grades for individual risk sets, as well as conduct a conditional density ratio test for the data to show that the conditional densities of student characteristics do not jump across the cutoff (Zimmerman, 2014). [5]

Additionally, it is important that the probability change of aid receipt at the threshold is significant. I plot the change in probability across the cutoff for both stipends and waivers in RSU in figure 5 and for UL in 6. I find that the first stage for waivers is large and statistically significant, whereas the jump in aid receipt probability for stipends is smaller. This difference aligns with the financial aid assignment mechanism, where for stipends GPA-rank is only part of the criteria.

[Figure 5 about here.]

[5] The other side of the self-selection risk is the negative selection effect wherein students who might have performed better might decrease their performance if they know the precise cutoff. The competitive nature of the aid reduces this risk, as students cannot precisely predict where the cutoff will be as it is based on the number of available seats, applicants, and their grades. This means that a student who might be well above the cutoff would still be unable to safely reduce their performance, as they would not be able to know beforehand where the cutoff will be.

[Figure 6 about here.]

Furthermore, to estimate the long-term effects of the reception of aid, I estimate a regression that compares individuals just above and just below the cutoff over time.

$$Y_{ijt} = \sum_{\tau=1}^{\bar{\tau}} [\beta_\tau Above_{ijt-\tau} + f_\tau(r_{ijt-\tau})] + \theta_j + \epsilon_{ij} \tag{2}$$

Where *Above* is an indicator variable equal to 1 if the student is above the cutoff and τ indexes periods after the competition, with the competititon for aid occuring in period 0.

This approach estimates the effect of reception of stipends irrespective of whether the student received stipends previously or will receive further stipends in the future. This is of economic significance, as it allows for a contextualized estimate wherein students applying for financial aid have histories and wherein aid can have a multiplying effect, begetting more aid in the future and larger effects.

However, this approach is not valid for estimating the effects of reception of aid for different students in different point in their studies. If the question faced by the policy-maker is when to best provide a limited amount of aid to a student with no other aid being available, then an approach that controls for a history of reception of financial aid is required.

3.1.2 Estimating the Dynamic Treatment Effects of Aid

Due to the competition repeating, aid eligibility also has a dynamic impact not only on future student outcomes, but also on future student aid eligibility. The effect on future eligibility ($\pi_{t,t+h}$) represents the average causal effect of crossing the eligibility threshold in semester t on the probability of crossing the threshold h semesters later. This parameter estimates the crowding-in effect of aid, i.e. how aid impacts reception of future aid in a competition. It is estimated from:

$$Above_{i,j,t+h} = \alpha + \pi_{t,t+h} Above_{ijt} + g(r_{ijt}) + \theta_{jt} + \nu_{ijt} \tag{3}$$

where $g(r_{ijt})$ is a flexible polynomial function of rank and θ_{jt} represents risk set fixed effects.

The dynamic nature of the aid allocation mechanism - where eligibility in semester t potentially improves student GPA, thereby increasing the probability of eligibility in semester $t+1$ (i.e., $\pi_{t,t+1} \neq 0$) - implies that the total effect ($\delta^T_{t,t+k}$) is a combination of the direct effect of semester t eligibility with indirect effects mediated through eligibility in intermediate semesters ($t+1, \ldots, t+k-1$). In other words, the total impact of aid operates not only through the reception of aid in a given semester, which then increases student performance, but also crowds in future aid, which further improves student performance. Thus, the total effect estimated in the previous section captures both the marginal effect of aid, and also the marginal effect of all future aid it crowds in as defined in equation 4.

$$\underbrace{\delta^T_{t,t+k}}_{\text{Total Effect}} = \underbrace{\delta^M_{t,t+k}}_{\text{Marginal Effect}} + \underbrace{\sum_{h=1}^{k} \left(\underbrace{\pi_{t,t+h}}_{\substack{\text{Effect on probability}\\ \text{of future eligibility}}} \times \underbrace{\delta^M_{t+h,t+k}}_{\substack{\text{Marginal Effect}\\ \text{in future semesters}}} \right)}_{\text{Crowding-in effect}} \tag{4}$$

To isolate the marginal effect ($\delta^M_{t,t+k}$) - the average causal effect of eligibility in semester t on the outcome at $t+k$, holding eligibility status in intermediate semesters constant - I employ methods from the dynamic RD literature to recursively decompose the impact into the marginal and crowd-in impact (Biasi et al., 2024; Cellini et al., 2010; Taylor, 2014). This approach isolates the direct influence pathway from $Above_{ijt}$ to $Y_{i,t+k}$ by mathematically removing the estimated influence flowing through intermediate eligibility changes:

$$\underbrace{\delta^M_{t,t+k}}_{\text{Marginal Effect}} = \underbrace{\delta^T_{t,t+k}}_{\text{Total Effect}} - \underbrace{\sum_{h=1}^{k} \left(\underbrace{\pi_{t,t+h}}_{\substack{\text{Effect on probability}\\ \text{of future eligibility}}} \times \underbrace{\delta^M_{t+h,t+k}}_{\substack{\text{Marginal Effect}\\ \text{in future semesters}}} \right)}_{\text{Crowding-in effect}} \tag{5}$$

where the sum of marginal effect equals the total effect, $\delta^M_{t+k,t+k} = \delta^T_{t+k,t+k}$ [6]. This recursive method constitutes my primary strategy for identifying the marginal impact of aid eligibility timing on both semester-level outcomes and final outcomes, such as graduation. It provides insights into the effectiveness of aid provision at different stages of a student's academic career, separate from its incentive effects on future aid receipt. A key assumption for this decomposition is the homogeneity of the dynamic parameters across different risk sets and student cohorts. For example, the crowding-in effect of aid in semester 2 is assumed to be stable over time across different cohorts.

For implementation, the required δ^T and π components are estimated simultaneously via a system of equations using stacked regressions to obtain the full variance-covariance matrix of the component estimates. Additionally, given that in this system there are two types of aid, I estimate π using the total aid amount that being above the cutoff crowds in - accounting for both aid received from waivers and from stipends. The marginal effects ($\delta^M_{t,t+k}$) are then computed recursively using Equation 5. Standard errors for $\delta^M_{t,t+k}$ are

[6] In effect, this means estimating the marginal effect for each and every future semester, so, for 3 semesters out, the estimating equation becomes

$$\begin{aligned}\delta^M_{t,t+3} = \delta^T_{t,t+3} - \pi_{t,t+1} \times & \underbrace{\left(\delta^T_{t+1,t+3} - \pi_{t+1,t+2}[\delta^T_{t+2,t+3} - (\pi_{t+2,t+3} \times \delta^T_{t+3,t+3})] - (\pi_{t+1,t+3} \times \delta^T_{t+3,t+3})\right)}_{\delta^M_{t+1,t+3}} \\ - \pi_{t,t+2} \times & \underbrace{\left(\delta^T_{t+2,t+3} - (\pi_{t+2,t+3} \times \delta^T_{t+3,t+3})\right)}_{\delta^M_{t+2,t+3}} \\ - \pi_{t,t+3} \times & \underbrace{\left(\delta^T_{t+3,t+3}\right)}_{\delta^M_{t+3,t+3}}\end{aligned}$$

calculated using the delta method and through a cluster bootstrap.

A key assumption underpinning the interpretation of $\delta^M_{t,t+k}$ as the marginal effect is that the contemporaneous effect of eligibility, $\delta^T_{t+h,t+h}$, is independent of prior eligibility history (i.e., eligibility in periods $t, \ldots, t+h-1$). This assumption is similar to the assumption of the regression discontinuity design, wherein aid eligibility at the cutoff, conditional on the running variable and risk set, is independent of all other outcomes.

4 Results: Static and Dynamic Impacts of Aid

4.1 The Static Impact of Financial Aid in a Competitive System

This section presents the overall impact of receiving financial aid (tuition waivers or stipends) on student effort and persistence throughout their studies, as estimated by a standard regression discontinuity design (Equation 1). These ”total effects” (δ^T) are of primary interest to policymakers as they reflect the consequences of awarding aid within the existing institutional framework, where aid is competitively reassigned each semester. Thus, these estimates inherently capture not only the immediate effect of aid but also any crowding in that current aid receipt provides for securing aid in future competitions.

4.1.1 Impact of Waiver Receipt

In this subsection, I analyze the effects of acquiring a waiver irrespective of previous and future history of reception of aid. This corresponds to the estimation of the treatment effects in equation 1 and provides estimates on the effect of acquiring a waiver for the next semester when the counterfactual is not acquiring it for the next semester. This is the margin under which many scholarships operate, as students have many opportunities for the acquisition of aid throughout their studies.

I now turn to estimating the impact of the reception of waivers. I first plot the impact of the provision of waivers in figure 7, and report the results in table 4

[Figure 7 about here.]

[Table 4 about here.]

I find that waiver reception increases GPA in the next semester by 0.43σ, persistence by 9.6pp, the probability of being on track[7] by 7.8, graduation by 11.9pp, and decreases the probability of dropping out by 14.9pp.

Rescaling these estimates to evaluate the impact of aid per €1'000 of aid prived, I find smaller effects. €1'000 of aid in waiver receipts increases GPA in the next semester by 0.121σ, persistence by 4.4pp, probability of being on track in the next semester by 3.6pp, graduation by 5pp, and decreases dropout by 6.5pp.

4.1.2 The Impact of Stipend Receipt

Results for RSU

I begin with a graphical representation of the changes in student outcomes across the threshold for a pooled RD across all programs, years, and semesters. I then continue with estimating the effects of the reception of aid on student outcomes.

[Figure 8 about here.]

Figure 8 shows the changes for student outcomes based on their rank distance from the person with the lowest GPA to have received aid in their risk set.

Table 5 reports the results from the estimates of the impact of aid receipt on the academic outcomes of students. What I find is that being eligible for stipends has much weaker effects than waivers, with students receiving stipends seeing considerable increases in their next-semester GPA of 0.3σ, but no effects on other student outcomes.

The results for stipends need to be considered in light of the results for individuals who achieve a stipend on top of receiving a waiver. The waivers provide stability and

[7] Being on track is defined as the student's semester number being the next number after their current semester. In Lavia, as in many countries in Europe, students choose to study in specific programs upon enrolment, which have a pre-defined course and semester sequence.

support for individuals, with the stipends additionally being able to elicit more performance from the students.

[Table 5 about here.]

Results for UL

I show the visual changes in the outcomes for students at the stipend threshold in UL in figure 9.

[Figure 9 about here.]

I find that being above the cutoff increases the probability of receiving a stipend by 96pp, and stipend receipt increases GPA by 0.213 σ in the next semester, increases graduation rates by 9pp, increased probability of being on track by 7.2pp, and increase on-time graduation rates by 5.8pp (table 6).

[Table 6 about here.]

The impacts per €1'000 of stipend aid are 0.428 σ on GPA, 17.5pp on the probability of being on track, and 14pp on graduation. These effects are larger than the impacts for waivers reported in table 4. The reason for such a differential effect could be that the University of Latvia does not have the same policy of guaranteeing waivers for all who got them upon enrollment, leading to a situation wherein students who receive stipends are able to successfully acquire more total aid in the future, as well as are less secure, leading to increased effort.

I hypothesize that the difference in impact comes from both UL having a much sharper cutoff and also from the fact that the student population at UL is more diverse and slightly lower achieving. Suggestive evidence from the previous section has shown that financial aid can be more effective at lower percentiles of achievement.

As an additional robustness test, I conduct this estimation using the estimation procedure suggested by Armstrong and Kolesár (2020) and find similar results (table 24 in the appendix).

4.1.3 Do the Effects Differ for Different Types of Students?

An essential question for further exploration of the effects of financial aid in the context of efficient program design is who is most impacted by it. Based on the theoretical model in the previous sections, I would expect to find that the individuals that most benefit from aid are those that have higher opportunity costs, as well as at lower levels of academic achievement and earlier in their studies. The reasoning behind this is that students at those margins are more credit-constrained and are thus more receptive to additional aid. Furthermore, given the compounding nature of aid, it is also likely that the earlier a student is provided financial aid, the more time the effect of aid has to compound.

The sequenced nature of aid, where the cutoff percentile changes from semester to semester, creates additional opportunities to investigate heterogeneities based on various financial aid program design characteristics and the characteristics of recipients. To explore these questions, I run further analysis by demographics, adding interaction terms with being above the cutoff in my regressions with various student demographics and characteristics of the provision of aid itself, such as its timing and the percentile at which the cutoff occurred. Table 7 reports the estimates for the heterogeneity of effects by student demographic factors in RSU.

[Table 7 about here.]

What I find is that generally the aid effectiveness does not vary by student demographics. I find evidence that stipends have stronger impact on student persistence and probability of being on track for students from the capital, but elicit less effort. Students from the capital are those that have completed their upper secondary education in the capital, where the campus of the university is located. Although the students are able to live with their parents, decreasing the costs, they are also more embedded in the economy and have more opportunities.

I argue that the answer lies in the role of liquidity. Students from the capital, who have better access to the local labor market, face a more acute day-to-day work-study conflict. Their challenge is not long-term solvency but immediate cash flow to manage weekly expenses. A monthly stipend directly addresses this problem by providing liquid, fungible cash that acts as a substitute for income from part-time work, thus buying back valuable time for academics. A waiver, in contrast, is an illiquid benefit. While it increases a student's net worth, it doesn't help them pay for transport or groceries this week. Because it fails to relax the immediate liquidity constraint driving the time-allocation decision, it has no impact on persistence for this group.

4.1.4 The Impact of Providing Aid as a Tournament

The way a competition is designed can both help and hurt student performance. On the positive side, tournaments motivate students in two main ways. First, winning increases the chance of winning again in the future, giving students a long-term reason to try hard. Second, these competitions are especially good at pushing the highest-performing students to work even harder. However, there is a downside. If students feel they have no chance of winning - for instance, if awards are only given to a tiny percentage of participants - they can become discouraged and perform worse. This is known as the "sore loser problem."

I estimate the impacts of financial aid by the characteristics of the risk set, at which percentile within the risk set the cutoff occurs, as well as how late throughout one's studies. Table 8 reports the results for RSU.

[Table 8 about here.]

I explore the option value of financial aid as described by the literature on performance tournaments by looking at the impact of financial aid at the end of one's studies when there is no future option for renewal. I find that there are considerably stronger effects on persistence for students in the first half[8] of their studies, with students in the first half receiving waivers having increased probability of persistence by 8.7pp, probability of being on track by 7.8pp, and decreased probability of dropout by 12pp. Similarly, students receiving waivers in the last year have smaller impacts on persistence and probability of being on track, and students receiving waivers in their last semesters seeing smaller impacts on their dropout and graduation rates.

Exploring heterogeneity by cutoff characteristics, I find evidence that financial aid is more effective at lower percentiles. For waivers, moving from the bottom to the top percentile of the cutoff decreases the impact on persistence by 31.5 percentage points and the probability of being on track by 32.3 percentage points. This implies that financial aid tournaments are more effective when the number of awards relative to the number of participants is larger, which would align with the theoretical predictions of individuals being incentivized by tournaments more if they believe that they have higher chances of winning.

Estimating the impacts for UL in table 9. I find similar results to the heterogeneity analysis in RSU - aid in the first half has a lower impact on GPA, but increased impact on persistence, and aid in the last semester has a smaller effect on graduation.

[Table 9 about here.]

I also explore the risk versus security effects of financial aid. The impact of the tournament can also vary by the intensity of competition. To estimate the impact of increased levels of competition and risk, I leverage several sources of variation.

First, I leverage the fact that waivers, if provided starting in the fall semester, are provided for the whole academic year or two semesters. Meaning that individuals who receive financial aid in the fall should feel less of a competitive effect in the first semester, as they have secured financial aid for the next semester irrespective of their performance.

[8] The reason that the first half is used rather than semesters is due to the fact that the university has very different program lengths, from 6-year programs for medical professionals to 2-year master's degrees

Second, I explore heterogeneity by the level of difficulty of the risk set. To do so, I use two different measures of difficulty - average GPA and average rate of being on track after the semester. Finally, I also leverage the fact that individuals who begin their studies with a tuition waiver have the waiver guaranteed for the rest of their studies. Thus, they are less at risk of losing both types of financial aid. I explore heterogeneity for the impact of stipends for this group. I report the results in 10.

[Table 10 about here.]

Generally, I find suggestive evidence that the impact of financial aid differs by the difficulty of the risk set. Risk sets with below median GPA see 0.339 σ larger impacts on their GPA in the next semester from waiver receipt, and increased persistence by 8.7pp. Risk sets with below median progression see similarly large waiver impacts on GPA of 0.389 σ, and 19.4pp impacts on persistence and 17.8pp impoacts on being on track. There are also security effects of aid, with students who have guaranteed aid having a higher impact on their next semester GPA of 0.198 σ.

Turning to estimating the heterogeneity by competitiveness for UL (table 11), I find that there are stronger security effects for stipends - stipends provided in the fall, when waivers are guaranteed, have increased impact on the probability of being on track by 6pp, graduation by 4.1pp and graduation on time by 5pp. However, the security effects do not elicit additional performance, having no impact on GPA. At the same time, I find that risk sets with below median GPA see similarly smaller effects on the persistence margin, decreasing graduation rates by 3.6pp. As opposed to RSU, I also find lower impacts on student performance as measured by GPA, with 0.14σ smaller impacts on GPA.

[Table 11 about here.]

4.2 Decomposing Dynamic Treatment Effects: Marginal Impacts, Crowding-In, and Optimal Timing

The previous section established the total impact of financial aid in a system where aid is reallocated based on ongoing performance. However, these total effects amalgamate the immediate, direct impact of receiving aid in a given semester with the indirect impact that arises because aid receipt can improve performance and thus increase the probability of receiving aid in subsequent semesters (a ”crowding-in” effect). This section aims to disentangle these components using the dynamic regression discontinuity framework. Specifically, I first examine the extent to which current aid receipt affects future aid probability ($\pi_{t,t+h}$). I then isolate the marginal impact of a single aid instance ($\delta^{M}_{t,t+k}$), holding future aid receipt constant, and explore its persistence. Finally, I investigate how the timing of aid (early vs. late in a student's academic career) affects long-term outcomes like graduation.

4.2.1 The Crowding-In Effect of Aid

I conduct further analysis to evaluate the effects of being offered aid on student outcomes in subsequent semesters. I use lagged values of the dependent variables to further explore the impact of the provision of aid in future semesters and report them in table 12 for waivers and table 13 for stipends.

[Table 12 about here.]

[Table 13 about here.]

I generally find that being eligible for either type of aid has considerable crowding-in effects on future aid. Being eligible for a waiver increases their probability of receiving a waiver for up to five semesters ahead, increasing their probability of receiving a waiver in five semesters by 14.3pp, and crowding in additional aid amounts. This effect is driven by individuals persisting through their studies, as being eligible for waiver increases persistence by 6.6pp after five semesters.

Similarly, stipend receipt also crows in additional stipends, with individuals after five semesters being 5pp more likely to hold a stipend. This effect, however, seems to be driven primarily by student effort, as stipend receipt increases future student GPA.

I find that UL has similar long-term impacts of aid eligibility, with students eligible for aid in the next semester seeing 9.7pp larger probabilities of reception of aid after 5 semesters, and increased GPA by 0.2σ after 4 semesters, increased probability of persistence by 1.1pp after 5 semesters, and drawing in additional €235 of aid in the subsequent semesters after acquiring €489 in aid in the next semester.

[Table 14 about here.]

These estimates correspond to the impact of aid eligibility in a system where aid is competitively reassigned and answer the question of ”what is the impact of being eligible for financial aid, given that it will be reassigned later?”. These effects include both the marginal effects of aid eligibility, such as increased performance in the semester of aid reception, and the indirect effects of aid eligibility, such as increased probability of aid eligibility in future semesters, which in turn increases performance in future semesters, and so on.

4.2.2 Intertemporal Impact of Aid: Early vs. Late Aid on Graduation

In this subsection, I explore whether the marginal impact of aid on long-term success, specifically graduation, differs depending on when it is received in a student's academic trajectory (e.g. early vs. late). I estimate the marginal impact of aid for different semesters.

To estimate the marginal impact of aid in any given semester, I first subset the data to take students who were undergraduates and who had a program length of 8 semesters (the most popular program length for undergraduate study), and who started and whose expected graduation falls within the timeframe of the data.

The goal of this subsetting is to ensure that the estimates are from individuals who are observed throughout the time period that I have data for and who have the same program length, to ensure that effects are not driven by attrition at the program level. In this section I report the estimates for waivers due to their impact, the results for stipends are available in the appendix. I report my estimates in figure 10

[Figure 10 about here.]

I find that the impact of waiver eligibility is higher in earlier semesters for most, with the effect only being statistically significant in the first year or first two semesters for increasing graduation and decreasing dropout.

The reason for this can also be found in the setting. In figure 11 I plot the probability of ever dropping out by individuals in each semester. This plots the probability of ever dropping out of the program for all individuals who have been observed in that semester. What I find is that the probability of dropping out is highest in the first semesters, after which it decreases, only jumping again at the final semester.

[Figure 11 about here.]

Additionally, I take the subset of individuals who were studying in 6-semester-long undergraduate programs (the most popular choice among students in UL), and conduct a decomposition into the marginal and indirect effects (figure 12). I find a similar pattern with the impact of aid on student persistence being highest in the earlier semesters, with the effect decreasing the second half of one's studies. At the same time, however, the impact on cumulative GPA stays strong throughout one's studies.

[Figure 12 about here.]

4.2.3 Marginal Effect of Aid vs Crowding-In

Using the recursive decomposition, I now turn to decomposing the total effect in every semester into the marginal effect (driven by aid eligibility in a given time) and the indirect effect (driven by future aid eligibility). I thus decompose the results in figure 10 into the total, marginal, and crowding in effects in figure 13 based on equation 3

[Figure 13 about here.]

Decomposing the effects into the marginal and crowd-in for UL (figure 14), I find that the marginal effect plays a larger role compared to the crowd-in effect for earlier semesters in UL. However, the crowd-in effect is most pronounced in the earliest semesters and decreasing over time.

[Figure 14 about here.]

Generally, what I find is that the considerable impact of aid in earlier semesters is affected by their crowding-in effect of future aid. Furthermore, the crowding-in effect is strongest in the first semesters, after which it dissipates.

4.3 Dynamic Complementarities in Financial Aid

The unique mechanism of aid allocation also allows further exploration of dynamic complementarities in aid. Based on the dynamic complementarity framework (Cunha & Heckman, 2007), individual investments in human capital are more effective for individuals with higher a priori levels of human capital. Based on this theoretical framework, providing financial aid might be more beneficial to individuals who already have higher levels of human capital, as they are able to produce additional human capital more efficiently.

To explore these mechanisms, I assume GPA as a proxy for human capital and use a two stage least squares approach, wherein I estimate the following equation:

$$CGPA_{i,t+1,j} = \beta_0 + \beta_1 CGPA_{itj} + \beta_2 Aid_{itj} + \beta_3 (CGPA_{itj} \times Aid_{itj}) + f(r_{itj}) + \theta_{tj} + \epsilon_{it} \quad (6)$$

Which is equation 2 with an additional interaction of aid and previous CGPA. And I instrument Aid_{itj} and $(CGPA_{itj} \times Aid_{itj})$ using the following equations:

$$\begin{aligned} Aid_{itj} = \pi_{10} + \pi_{11} Above_{itj} + \pi_{12}(CGPA_{itj} \times Above_{itj}) \\ + \pi_{13} CGPA_{itj} + f(r_{itj}) + \theta_{tj} + v_{1it} \end{aligned} \tag{7}$$

$$\begin{aligned} (CGPA_{itj} \times Aid_{itj}) = \pi_{20} + \pi_{21} Above_{itj} + \pi_{22}(CGPA_{itj} \times Above_{itj}) \\ + \pi_{23} CGPA_{itj} + f(r_{itj}) + \theta_{tj} + v_{2it} \end{aligned} \tag{8}$$

This approach approach allows me to use the plausibly exogenous variation caused by the jump in aid receipt probability upon crossing the threshold to partial out the impact of aid specifically for individuals with different previous levels of human capital. The coefficient on beta 3 in the second stage equation would capture the effects of dynamic complementarity in skills and provides evidence on whether the investments in the skills of individuals more effective for individuals with higher skills a prior. I report the results in table 15 for waivers and stipends in RSU, and in table 16 for stipends in UL.

[Table 15 about here.]

[Table 16 about here.]

My results show strong support for the theory of dynamic complementarity, where investment is most effective for those with higher initial skills. The effect of aid on future academic performance is significantly magnified for students with higher existing levels of human capital. Specifically, at RSU, a one-standard-deviation increase in a student's current CGPA amplifies the positive effect of waivers on next-semester's CGPA by 0.12σ and the effect of stipends by 0.04σ. This finding is directionally supported by evidence from UL, where the same increase in prior CGPA is associated with a 0.02σ larger impact for stipends.

5 Discussion and Conclusion

In this paper, I estimate the effect of financial aid that is competitively reassigned each semester based on student merit. Using a unique institutional setting in Latvia where tuition waivers and stipends are the only form of aid available to the vast majority of students, I provide causal evidence on the dynamic effects of these programs on student success.

I find that both tuition waivers and stipends have a significant positive impact on student outcomes, increasing their grades, persistence, and graduation rates. The total effects are large: receiving a tuition waiver increases a student's probability of graduation by 11.9 percentage points and their persistence into the next semester by 9.6 percentage points.

These estimated impacts are at the upper end of the range reported in prior financial aid evaluations, which is likely driven by the tournament-style incentives and the lack of alternative aid sources in this setting. The robustness of these findings is enhanced by the fixed curriculum, which limits students' ability to game the system through course selection and provides a cleaner measure of aid's impact on academic effort.

The central contribution of this paper is the identification of a powerful and persistent crowding-in effect, where receiving aid in one semester significantly increases the probability of receiving it in subsequent semesters. To understand the importance of this mechanism, I decompose the total effect of aid into its direct marginal impact and its indirect crowding-in impact. This analysis reveals that the crowding-in of future aid is a substantial driver of the large long-term benefits, particularly for aid awarded to students earlier in their studies. This provides direct evidence for a form of dynamic complementarity where an initial aid award helps set students on a durably different and more successful academic trajectory.

While the baseline decomposition model assumes homogeneity for tractability, I explicitly test this assumption in my heterogeneity analyses. The findings in tables 7 and 10 show that the impact of aid and its dynamic components do in fact vary with the competitiveness and timing of the tournament. This suggests that the crowding-in effect is likely strongest in less competitive risk sets and earlier in a student's career.

Diving deeper into the mechanisms of skill formation, this paper provides a structural test of the interaction between existing skills and financial aid investment. The results provide evidence of dynamic complementarity, where investment is most effective for those with higher initial skills. The effect of aid on future academic performance is significantly magnified for students with higher existing levels of human capital. Specifically, at RSU, a one-standard-deviation increase in a student's current CGPA amplifies the positive effect of waivers on next-semester's CGPA by 0.124σ and the effect of stipends by 0.04σ. This finding is directionally supported by evidence from UL, where the same increase in prior CGPA is associated with a 0.02σ larger impact for stipends. This presents a key consideration for policymakers, highlighting that targeting aid towards high-performing students may be an effective strategy for maximizing human capital investment, as the aid not only rewards but actively multiplies their success.

Finally, the findings from Latvia's unique setting offer several lessons for policy design in more complex aid environments. The power of the crowding-in effect demonstrates that policymakers should evaluate aid not as a series of one-off grants, but as a system that can create positive feedback loops over a student's entire academic career. In multi-program systems like that of the United States, understanding how different grants and scholarships interact to either promote or hinder these dynamic complementarities is a critical area for future research. The results also highlight a key tension in merit aid design: while tournaments effectively incentivize effort, there is a trade-off between the intensive and extensive margin of providing aid - tuition waivers can ensure that more students remain enrolled and graduate, while additional stipends on top of waivers can elicit more effort.

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Appendices

A Density Tests for Sparse Data

The main assumption behind an RD design is that student potential outcomes are continuous across the cutoff, i.e. that the only student characteristics that suddenly change across the cutoff are their probability of treatment receipt. To support this assumption, there are multiple tests, with the two most prominent being a balance test of student characteristics (reported in table 3), and a density test of whether there is differential heaping around the cutoff (Cattaneo et al., 2020; McCrary, 2008).

The core concern that these tests address is whether students are able to select into treatment or not. If students are able to select into treatment, then the RD estimator will be biased. The balance test looks at observable characteristics and sees whether those change discontinuously, and the density test looks at the heaping around the cutoff to see if people are differentially selecting into treatment, which would show up as a large number of individuals being just below or above the cutoff.

The way my data is constructed is that students take courses in a semester that can only take on full values ranging from 1 to 10, with the semester grade being the average of these values. Because of the way the data is generated, regular density tests would yield incorrect inference, because the discontinuous distributions could just as well be attributed to the data-generating process as to student selection into treatment.

The tournament-style provision of financial aid means that for students to be able to select into treatment, they would need to push another student out, which would inherently balance the densities across the cutoff. However, the null hypothesis of no discontinuity is rejected by the density tests due to the data-generating process.

I plot the 10 largest risk sets for stipends (figure 15) and waivers (figure 16). Thus, when looking at each risk set individually and at the raw scores, I do not find evidence of individuals being able to precisely manipulate their GPA and self-select into aid reception.

[Figure 15 about here.]

[Figure 16 about here.]

To empirically estimate the degree to which there could be selective discontinuities across the cutoff, I use two different tests - a test for the discontinuity in the ratios of the conditional densities to the unconditional density (Zimmerman, 2014), and a permutation test. The permutation test is reported in the main section of the paper.

The conditional density ratio test looks at the conditional distribution of student densities across the cutoffs to see if they are continuous across the cutoff. The test looks at not only the continuous distribution of standardized grades across the cutoff, but of student conditional densities based on characteristics that impact their academic outcomes. I report the results of this test in figure 17 for three different groups - male students, students from minority-language-of-instruction schools, and students from Riga. I find that each of the density ratios is continuous around the cutoff value.

[Figure 17 about here.]

B Robustness Tests

I also conduct robustness tests by varying the bandwidth (tables 18 and 17). I find that my results are robust to different bandwidth specifications.

[Table 17 about here.]

[Table 18 about here.]

I also conduct robustness tests by varying the point at which the cutoff occurs and similarly find that centering the cutoff at 0 provides the strongest predicted probability of treatment.

[Table 19 about here.]

I also conduct these tests for UL and find that the results are also robust.

[Table 20 about here.]

[Table 21 about here.]

Finally, I also use the estimation approach by Armstrong and Kolesár (2020) to conduct tests using their suggested RD estimation approach. I let the bandwidth vary for each outcome variable and aid type and estimate the balance table and reduced form estimate tables in tables 22, 23, 24, and 25. I generally find broadly similar outcomes as with my main specification.

[Table 22 about here.]

[Table 23 about here.]

[Table 24 about here.]

[Table 25 about here.]

C The One-Step Dynamic RD Method

As a complementary approach, I also adapt the "one-step" dynamic RD estimator (Cellini et al., 2010) to estimate the dynamic impact profile of past eligibility events on current outcomes. The difference between the one-step estimate and the recursive estimate would be that the recursive estimate described in the previous section is that the recursive estimate estimates the marginal contribution of aid in different semesters on a student's outcome.

The one-step estimator, on the other hand, estimates the impact of aid reception after time t, without being able to decompose the impact into direct and indirect effects. Another way to think about it would be to think of the one-step estimator as estimating the decay pattern of financial aid, while holding aid history constant.

The benefits of the one-step dynamic RD approach is that it uses the full history of observations for each individual, thus increasing statistical power. It also estimates the impact of financial aid over time. Using this method, I estimate the following model using panel data:

$$Y_{ijt} = \beta_0 Above_{ijt} + f_0(r_{ijt}) + \sum_{\tau=1}^{\bar{\tau}} \left[\beta_\tau Above_{ijt-\tau} + f_\tau(r_{ijt-\tau})\right] + \lambda_j + \omega_t + \epsilon_{ijt} \tag{9}$$

where $Above_{ijt-\tau}$ indicates eligibility status τ semesters prior, $f_\tau(r_{ijt-\tau})$ is a flexible function of the rank from τ semesters prior, λ_j represents risk-set fixed effects, and ω_t denotes semester fixed effects. The coefficients β_τ estimate the average effect for individuals above the cutoff τ periods in the past on the current outcome Y_{ijt}, conditional on the history of eligibility and rank included in the model. Plotting β_τ against the lag τ directly illustrates the impact over time, or decay pattern, of initial eligibility. While estimating a different parameter than the marginal effect $\delta^M_{t,t+k}$ from the recursive method, this approach provides valuable insights into effect duration and will offer greater statistical precision for estimating the persistence pattern itself, as it will leverage the full dataset.

Additionally, for the one-step estimator I use the same subset of individuals as I use for the dynamic decomposition - people in undergraduate programs with a program length of 8 semesters.

I find that even when controlling for aid history, financial aid in past semesters has strong effects on outcomes in the current semesters, with waivers having impact two semesters ahead (figure 18) and stipends having extensive and pronounced impact on GPA throughout the observed history (figure 19)

[Figure 18 about here.]

[Figure 19 about here.]

Additionally, I explore the heterogeneity of financial aid impact decay by conditions of the competition. As waivers received in the fall are guaranteed for two semesters, I study how financial aid receipt in the fall compared to the spring impacts the effects over time in figures 19 and 21. I generally do not find differential impacts of having more stable waivers for students.

[Figure 20 about here.]

[Figure 21 about here.]

Finally, given that some students have guaranteed waivers throughout their studies if they received a waiver upon enrolment, I conduct a heterogeneity analysis for stipend impacts by interacting stipend aid receipt with whether the student had a guaranteed waiver or not (figure 22).

I find that individuals who received stipends while in guaranteed waiver seats see smaller impacts on their GPA over time, potentially driven by the security effect of having a guaranteed tuition waiver. This heterogeneity only applies to CGPA, with no significant difference for other variables such as dropout, graduation or persistence, implying that the main margin that stipends motivate students in RSU is through increased academic performance, but not affecting their academic progression.

[Figure 22 about here.]

D Additional Results for UL

This section reports additional regression discontinuity results for the University of Latvia, which also provides state-funded stipends, but only uses GPA for their allocation (as opposed to GPA and scientific performance for RSU).

E Dynamic Decomposition Results for Stipends in RSU

[Figure 23 about here.]

[Figure 24 about here.]

[Figure 25 about here.]

List of Figures

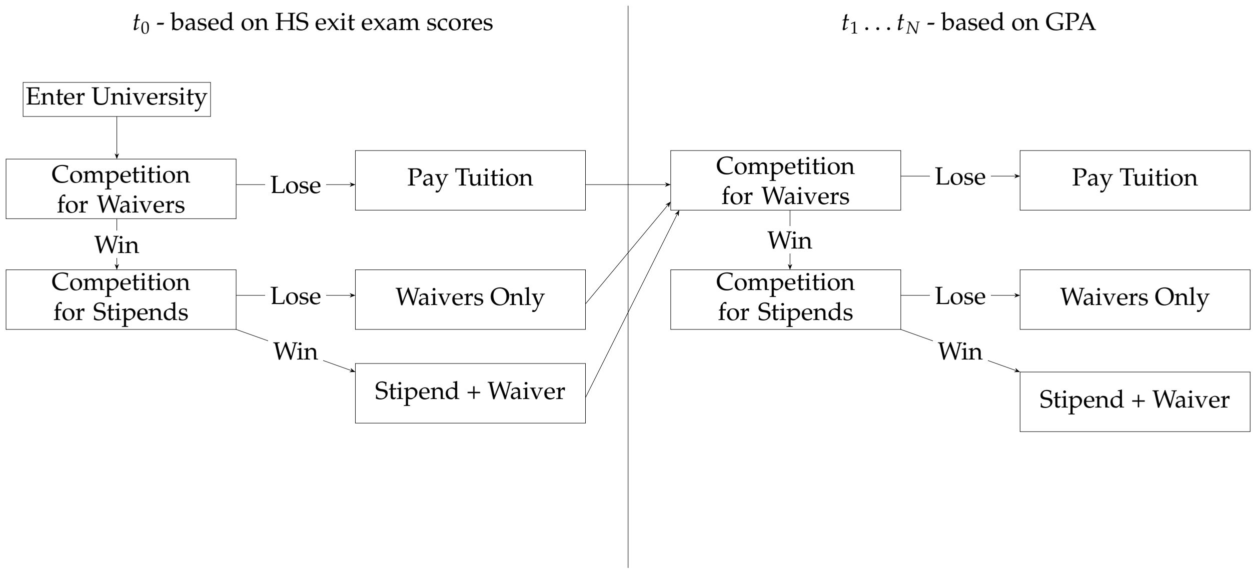
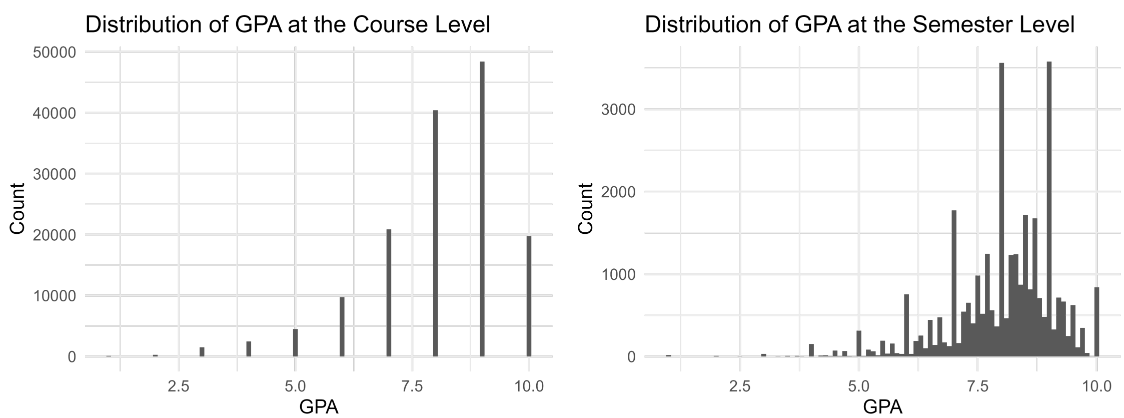


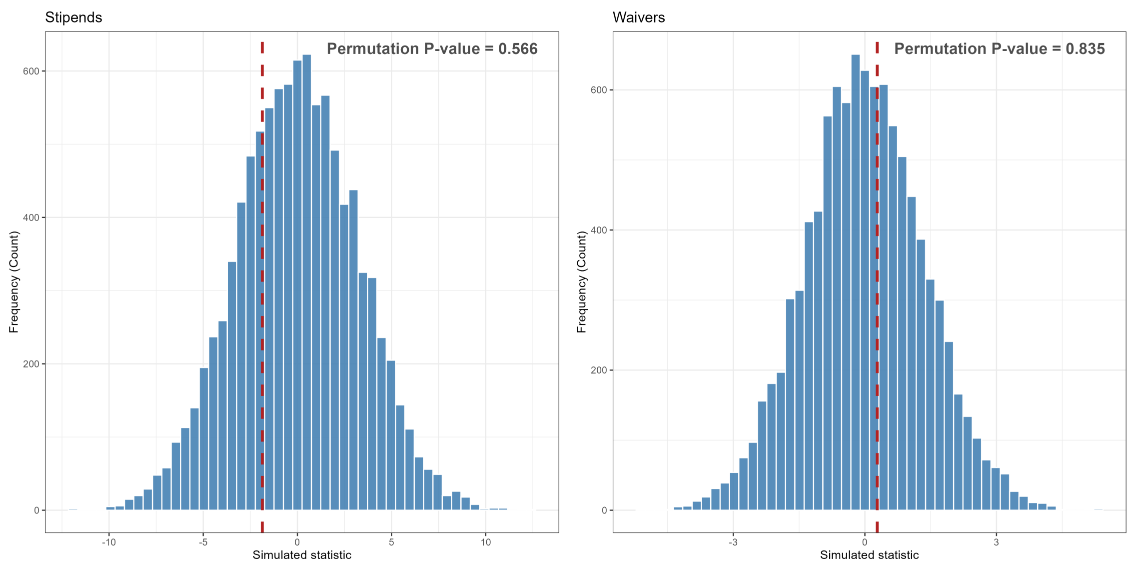
Figure 1: The Student Financial Aid Assignment Mechanism in a Dual-Track Tuition Model

Figure 2: Raw GPA at the Course and Semester Levels



Notes: The figure on the left shows the raw GPA that students acquire at the course level, and the figure on the right shows the average semester GPA that is used to assign financial aid.

Figure 3: Distribution of Density Test Values for RSU



Notes: These figures show the distribution of density test values for the dataset wherein the cutoff values are randomly reassigned within risk sets. Line indicates density test value for observed data.

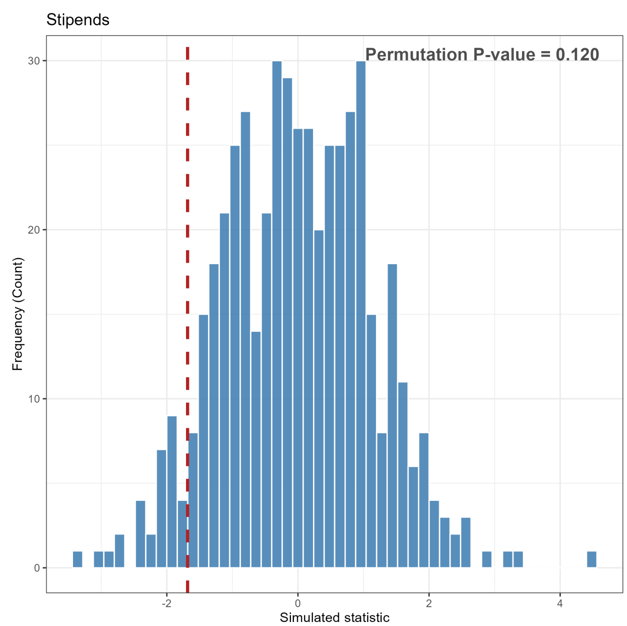
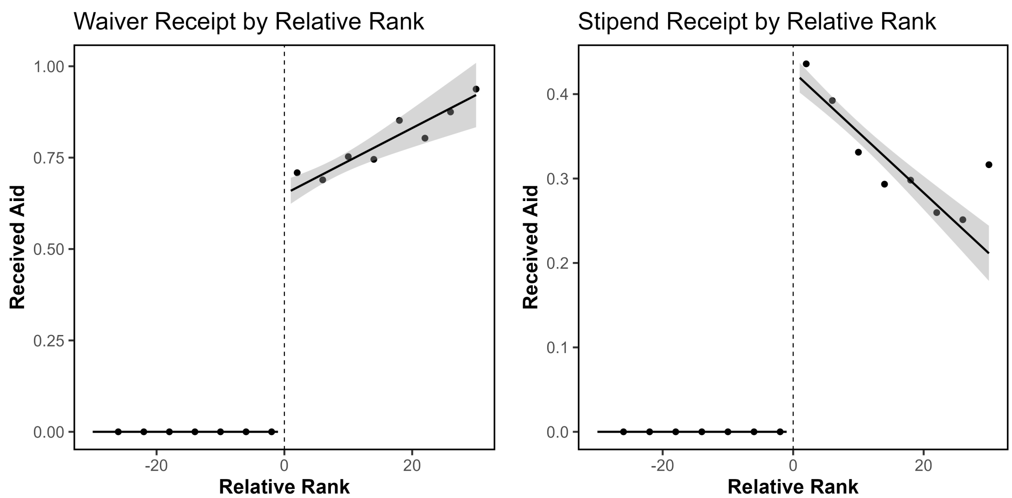


Figure 4: Distribution of Density Test Values for UL

Figure 5: Change in Aid Receipt Probability for RSU



Notes: These figures show the change in student probability of aid receipt based on rank distance from the individual with the smallest GPA to have received a given type of financial aid.

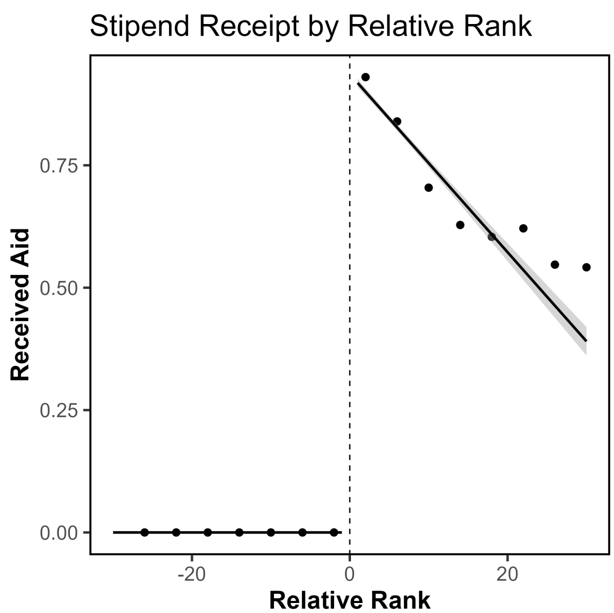
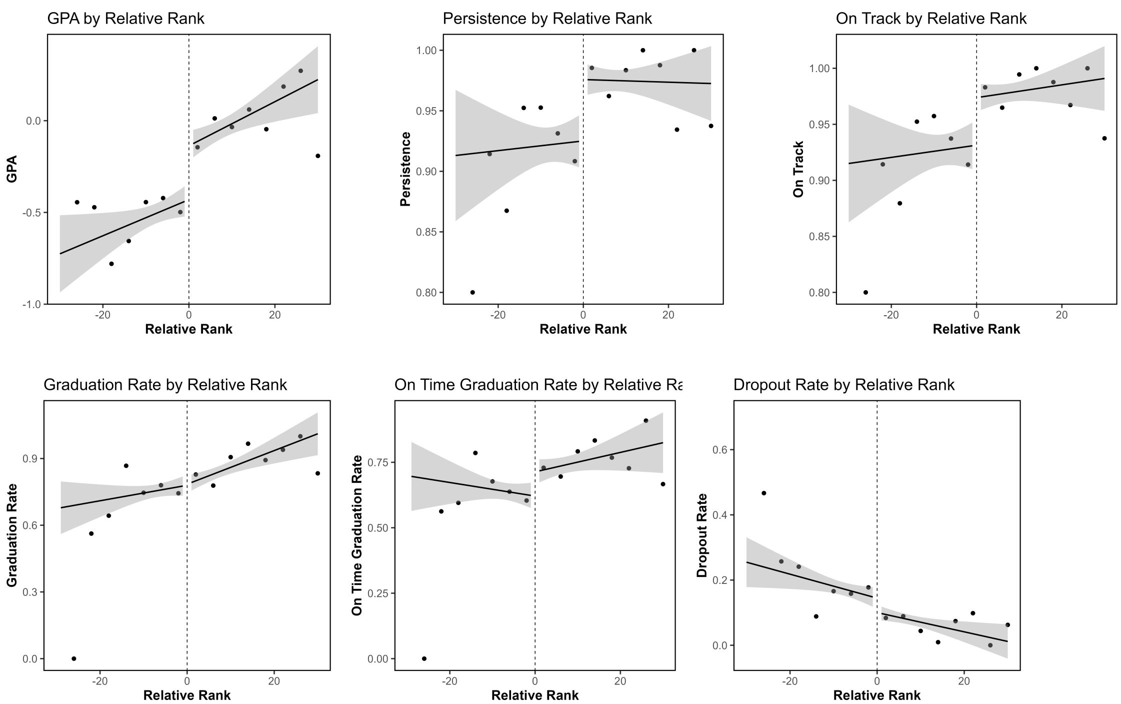


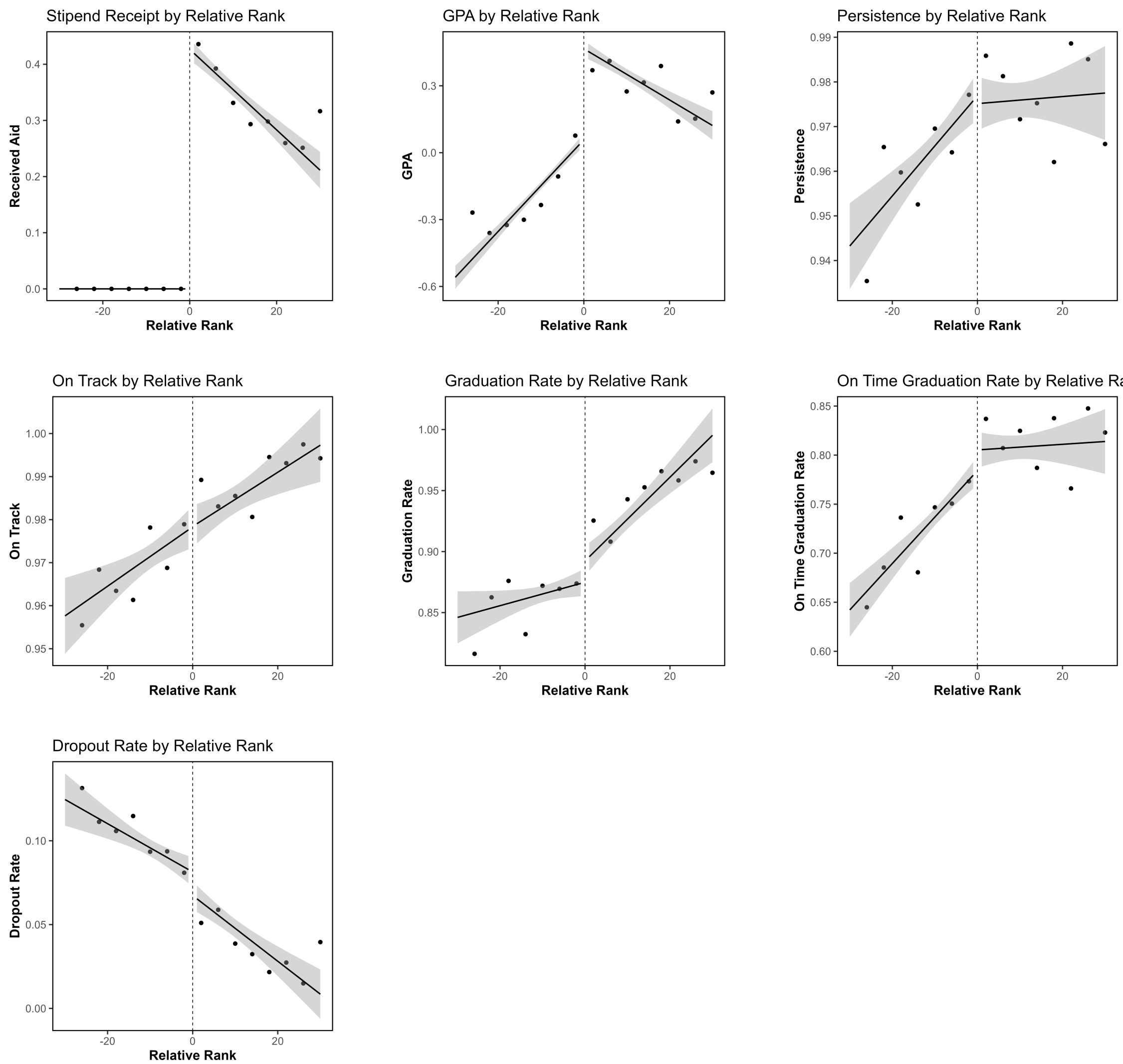
Figure 6: Change in Aid Receipt Probability in UL

Figure 7: Student Outcomes Below and Above the Tuition Waiver Reception Threshold



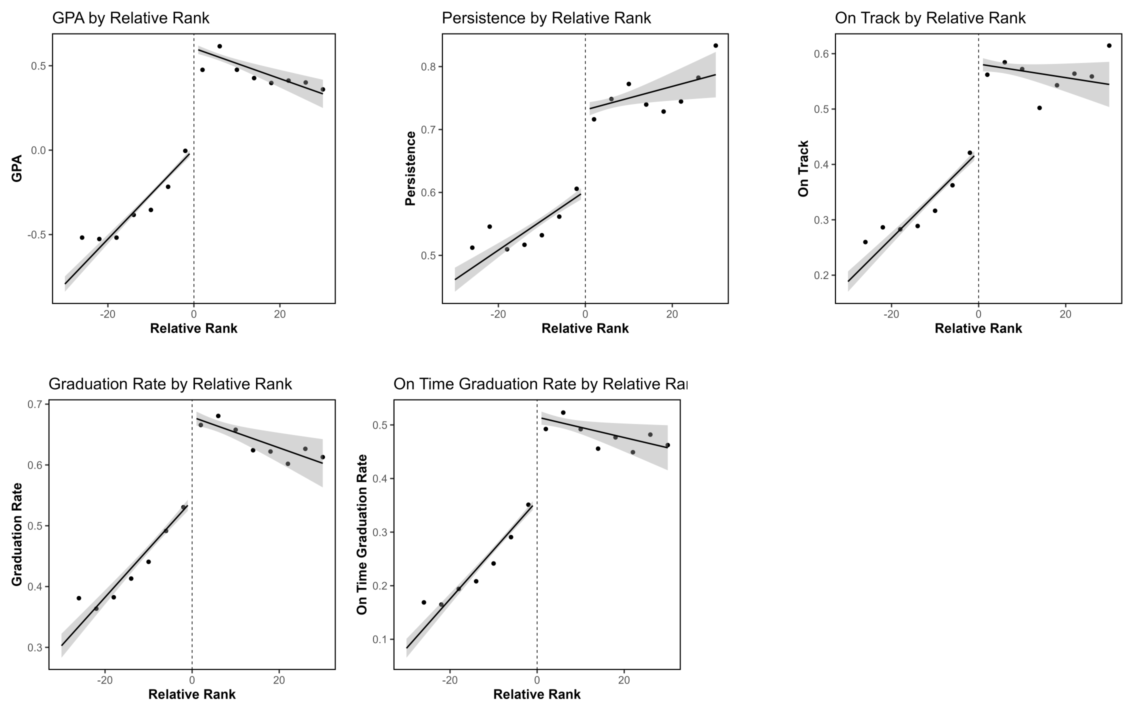
Notes: These figures show the change in student outcomes based on distance from the rank cutoff for reception of tuition waivers. The lines are fitted values. The points are average values based on the data.

Figure 8: Student Outcomes Below and Above the Stipend Reception Threshold for RSU



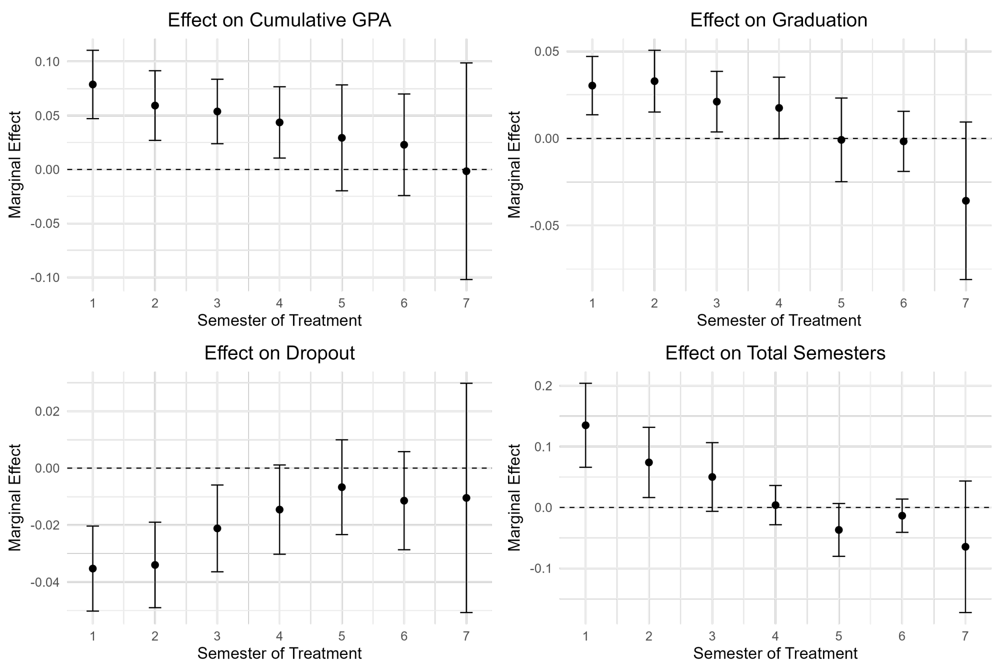
Notes: These figures show the change in student outcomes based on distance from the rank cutoff for reception of stipends. The lines are fitted values. The points are average values based on the data.

Figure 9: Student Outcomes Below and Above the Stipend Reception Threshold for UL



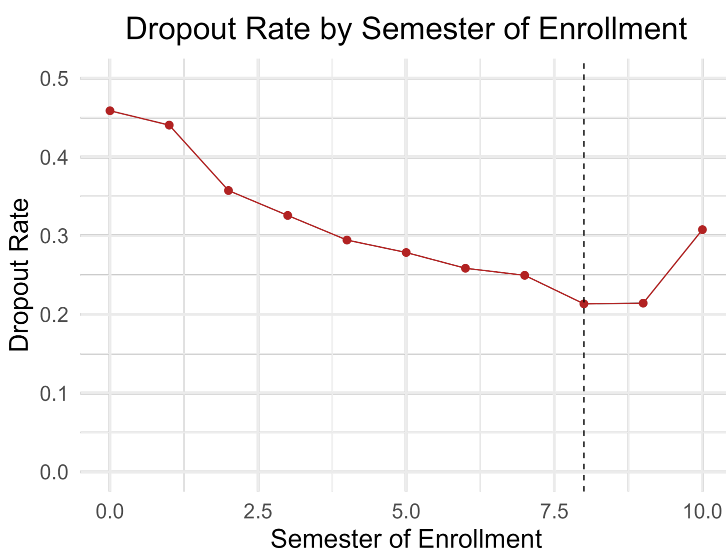
Notes: These figures show the change in student outcomes based on distance from the rank cutoff for the reception of stipends. The lines are fitted values based on the main specification with risk-set fixed effects. The points are average values based on the data.

Figure 10: Marginal Effect of Waiver Eligibility at a Given Semester



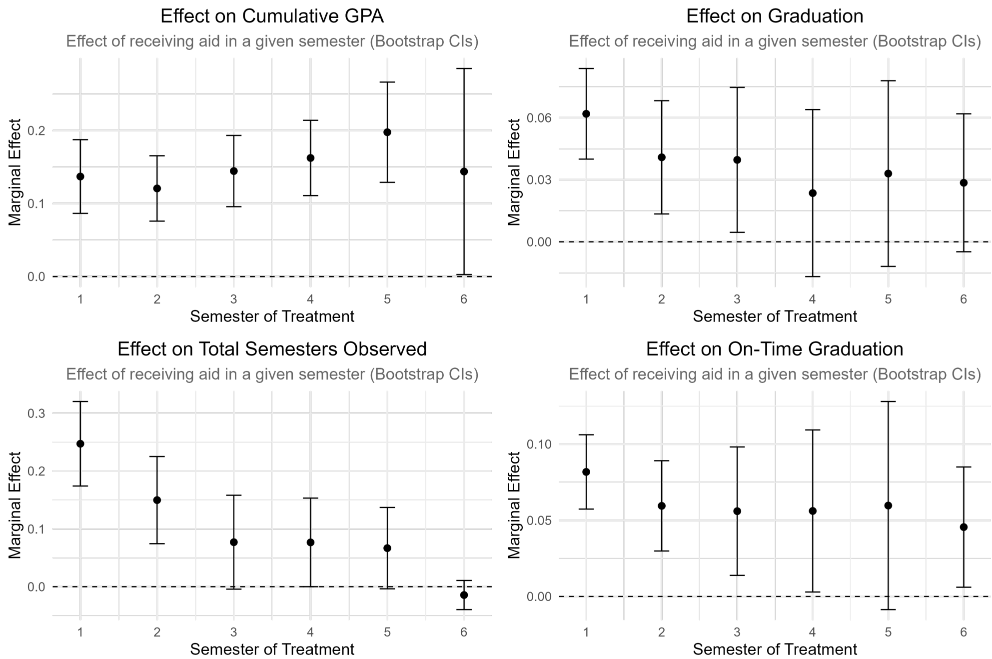
Notes: The figures report the estimates from an indirect regression discontinuity decomposition on the marginal impact of being eligible for waivers and total received aid in any given semester on a range of student outcomes - cumulative GPA, graduation, ever dropping out, and the total number of semesters that a student is enrolled for. The estimation is done on students who start during the time period and whose expected graduation falls before the data horizon, and whose program length is 8 semesters. Standard errors acquired by conducting a cluster bootstrap.

Figure 11: Probability of Dropping out by Semester for Undergraduates With a Program Duration of 4 years



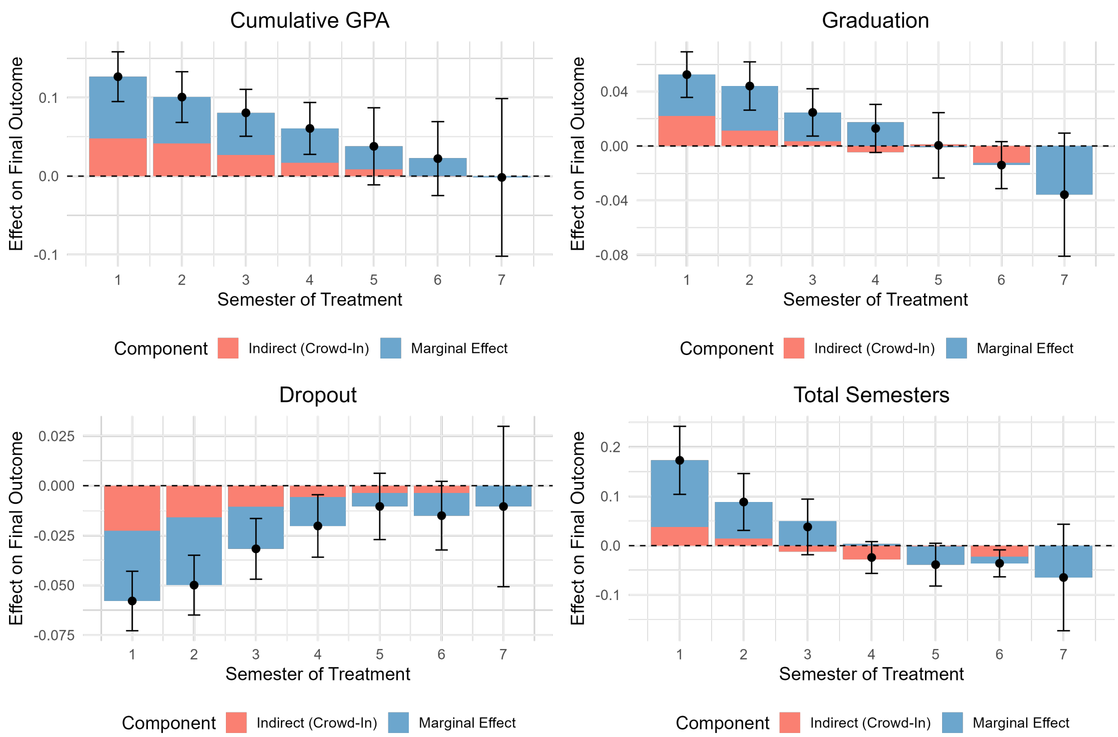
Notes: This figure reports the probability that an individual observed in a given semester would drop out at any future time for a subset of individuals who are studying in 4-year undergraduate programs. Dashed line indicates semester 8 - the final semester in one's studies for this subgroup.

Figure 12: Marginal Effect of Stipend Eligibility at a Given Semester with Bootstrap SEs for UL



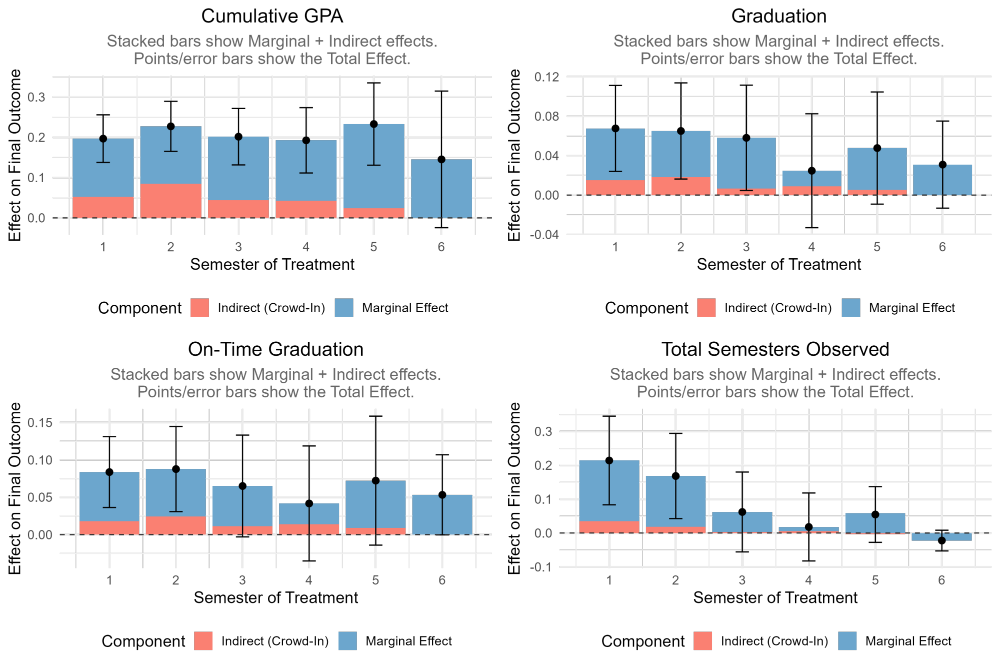
Notes: The figures report the estimates from an indirect regression discontinuity decomposition on the marginal impact of being eligible for stipends in any given semester on a range of student outcomes. The estimation is done on students who start during the time period and whose expected graduation falls before the data horizon, and whose program length is 6 semesters.

Figure 13: Decomposition of Impact of Waiver Eligibility



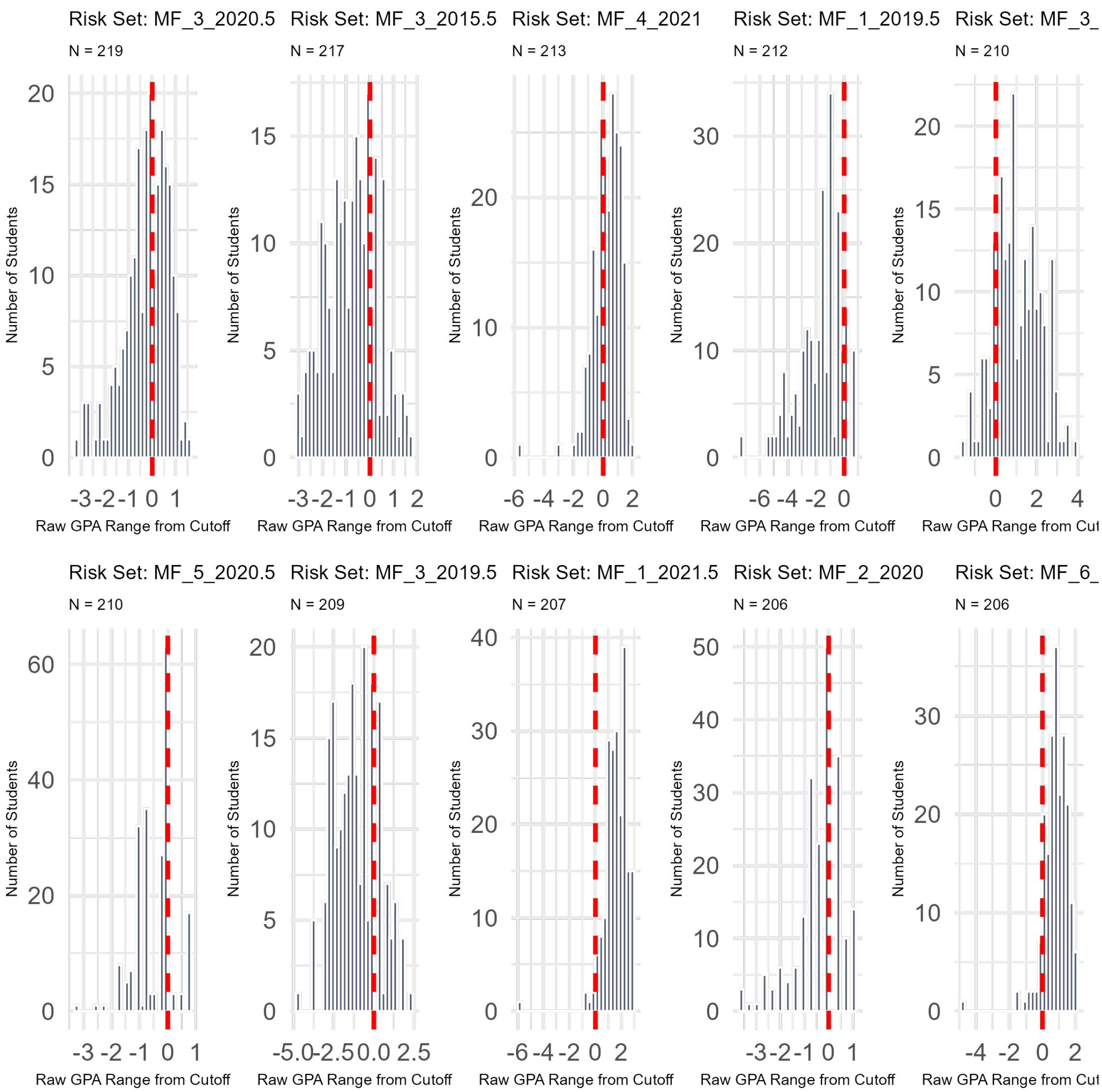
Notes: The figures report the estimates from an indirect regression discontinuity decomposition on the marginal, indirect, and total impact of being eligible for waivers in any given semester on a range of student outcomes. The estimation is done on students who start during the time period and whose expected graduation falls before the data horizon, and whose program length is 8 semesters. Standard errors acquired using the delta method.

Figure 14: Decomposition of Impact of Stipend Eligibility for UL



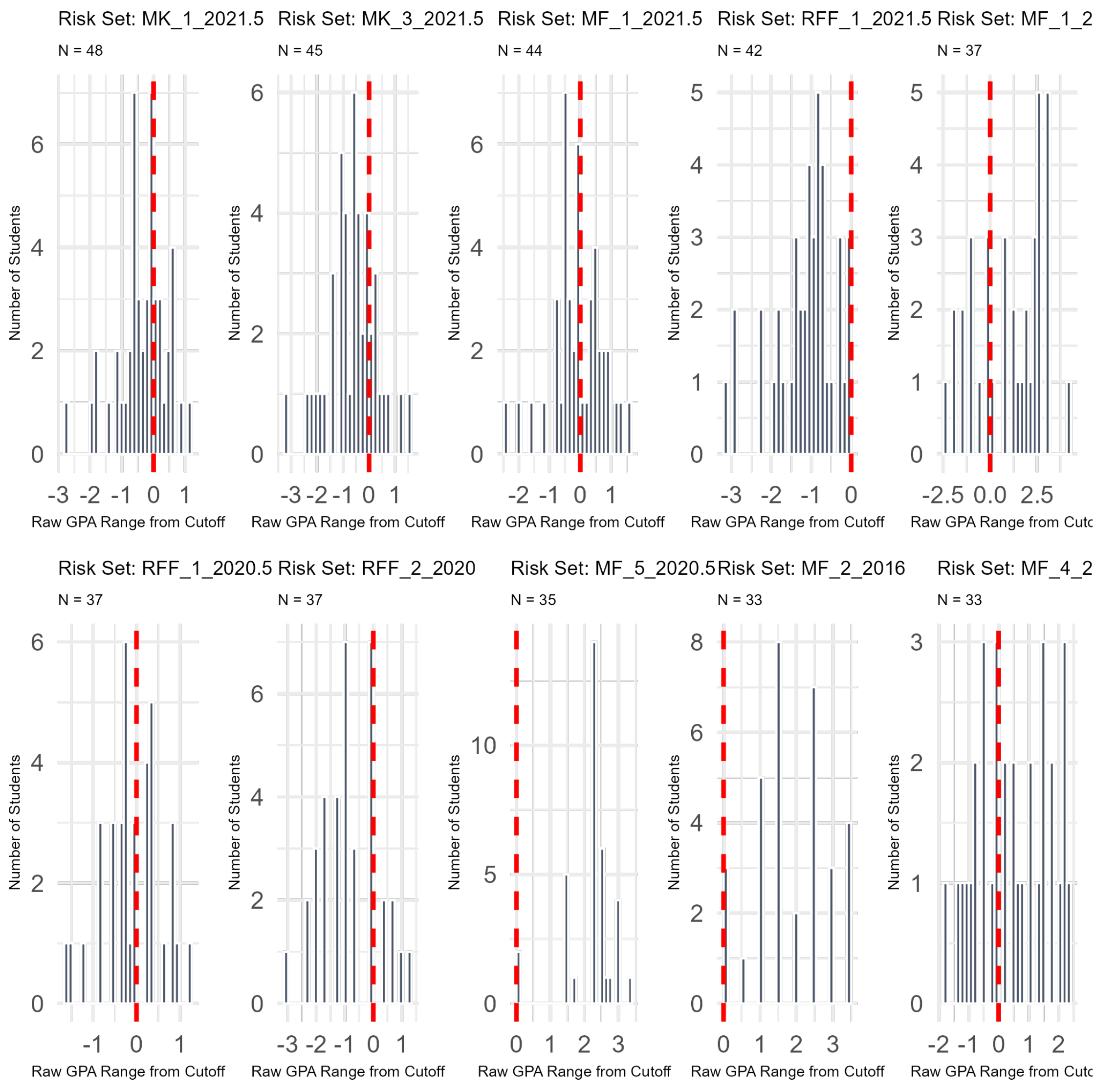
Notes: The figures report the estimates from an indirect regression discontinuity decomposition on the marginal, indirect, and total impact of being eligible for stipends in any given semester on a range of student outcomes. The estimation is done on students who start during the time period and whose expected graduation falls before the data horizon, and whose program length is 8 semesters.

Figure 15: Density of Raw GPA Scores for 10 Largest Stipend Risk Sets



Notes: These plots represent the density of the raw GPA of individuals in the 10 largest risk sets. The GPA has been recentered so that 0 is the cutoff at which the individual with the lowest GPA received aid.

Figure 16: Density of Raw GPA Scores for 10 Largest Waiver Risk Sets



Notes: These plots represent the density of the raw GPA of individuals in the 10 largest risk sets. The GPA has been recentered so that 0 is the cutoff at which the individual with the lowest GPA received aid.

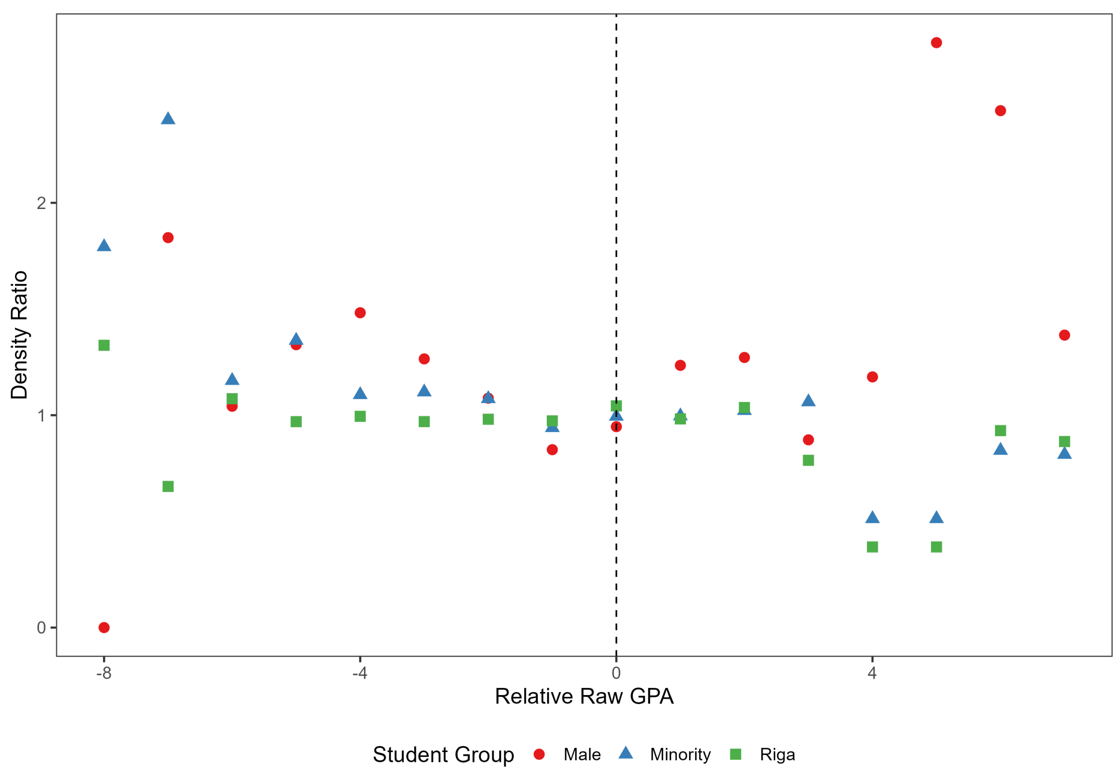
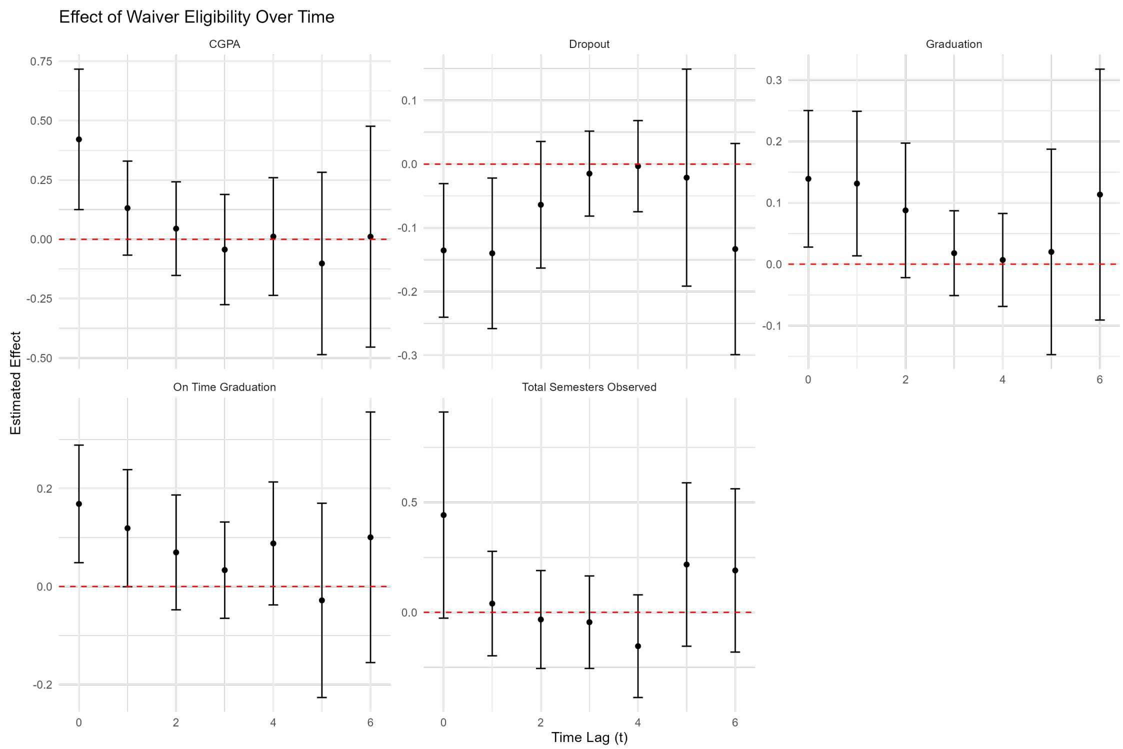


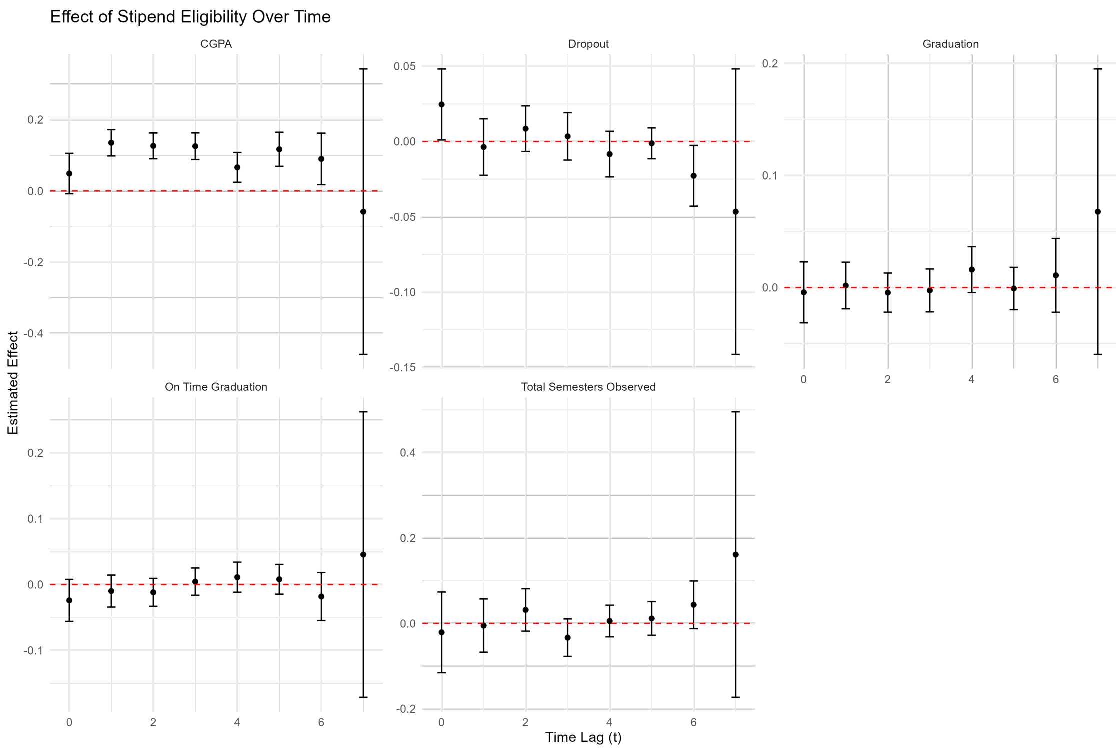
Figure 17: Conditional Density Ratio Test

Figure 18: One-shot estimator results for waivers



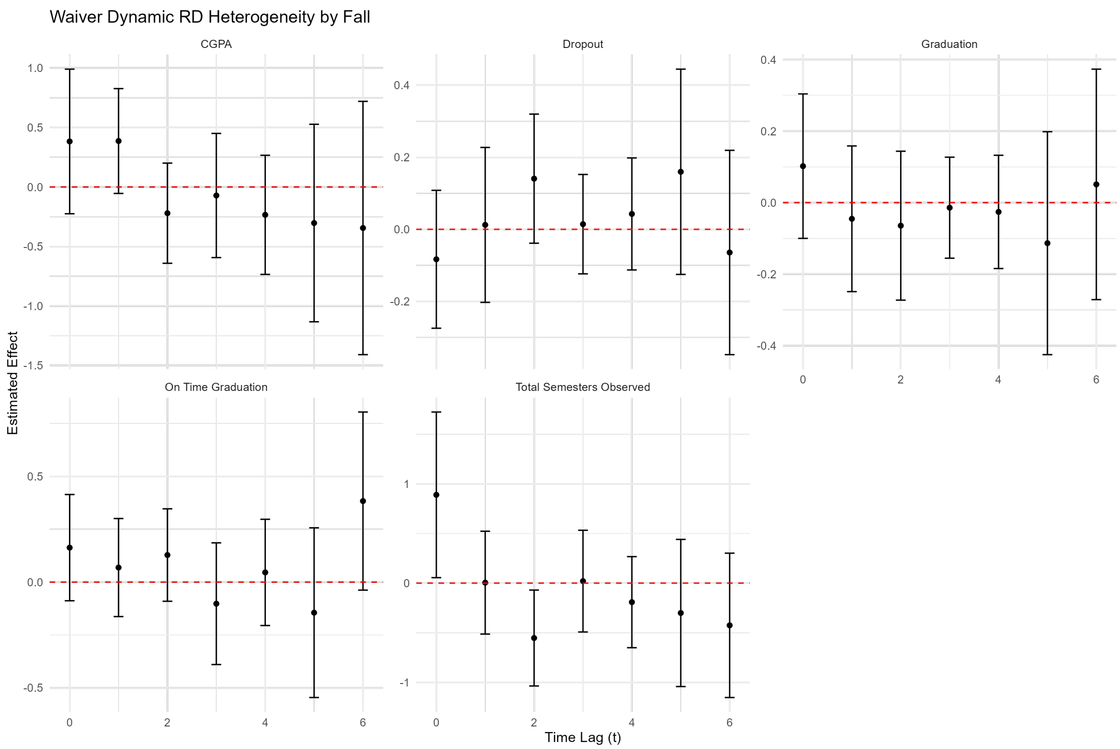
Notes: These figures show the results of the one-shot estimator for waivers for undergraduate students with a program length of 8 semesters. The figures report the estimates on waiver receipt in past on current grades, with the x axis reporting the lag.

Figure 19: One-shot estimator results for stipends



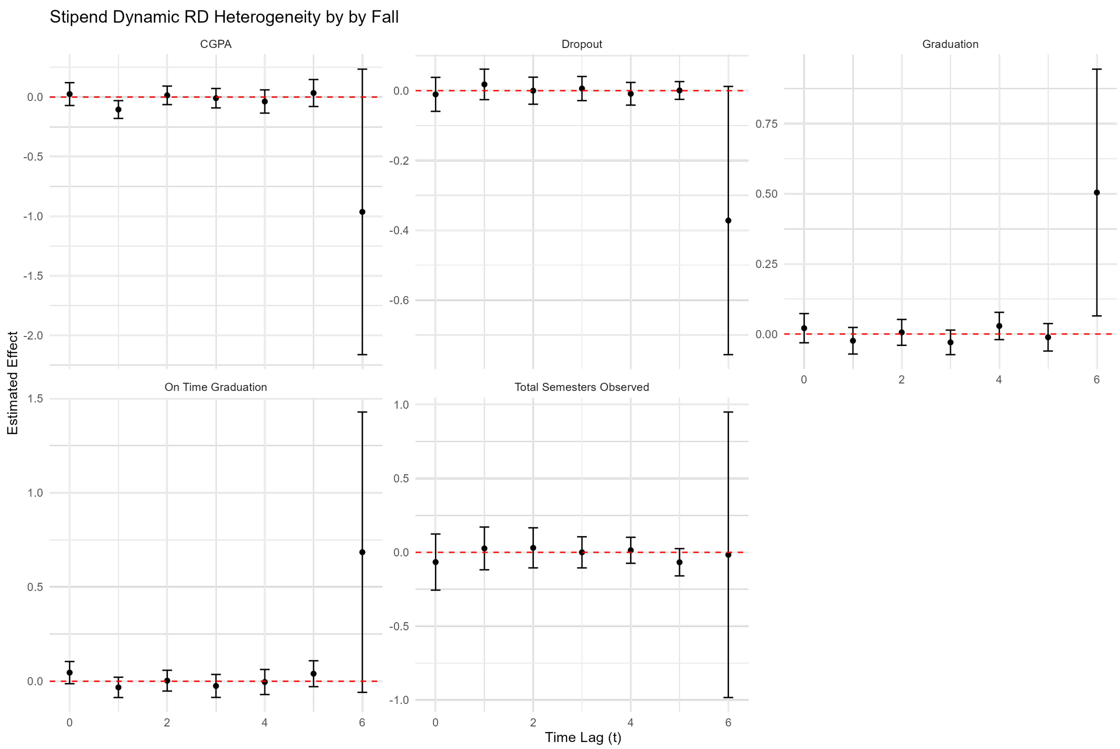
Notes: These figures show the results of the one-shot estimator for stipends for undergraduate students with a program length of 8 semesters. The figures report the estimates on stipend receipt in past on current grades, with the x axis reporting the lag.

Figure 20: One-shot estimator heterogeneity for waivers in the fall



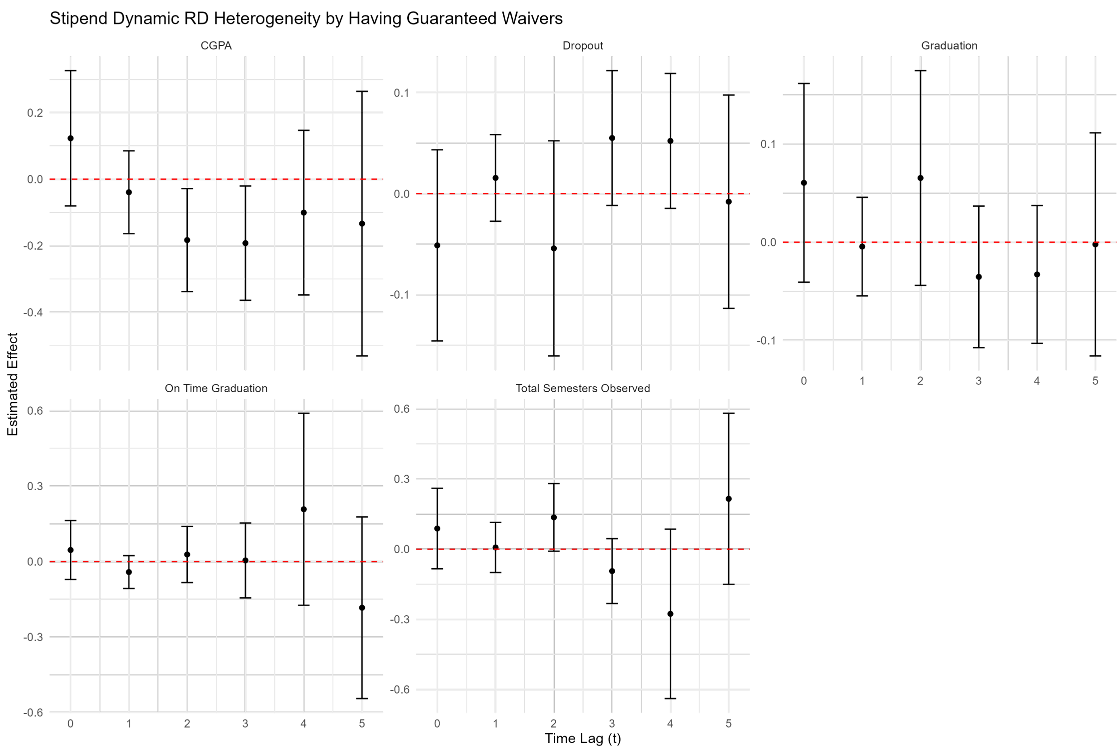
Notes: These figures show the results of the interaction term between past aid and receiving it in the fall for the one-shot estimator for waivers for undergraduate students with a program length of 8 semesters. The figures report the estimates on waiver receipt in past on current grades, with the x axis reporting the lag.

Figure 21: One-shot estimator heterogeneity for stipends in the fall



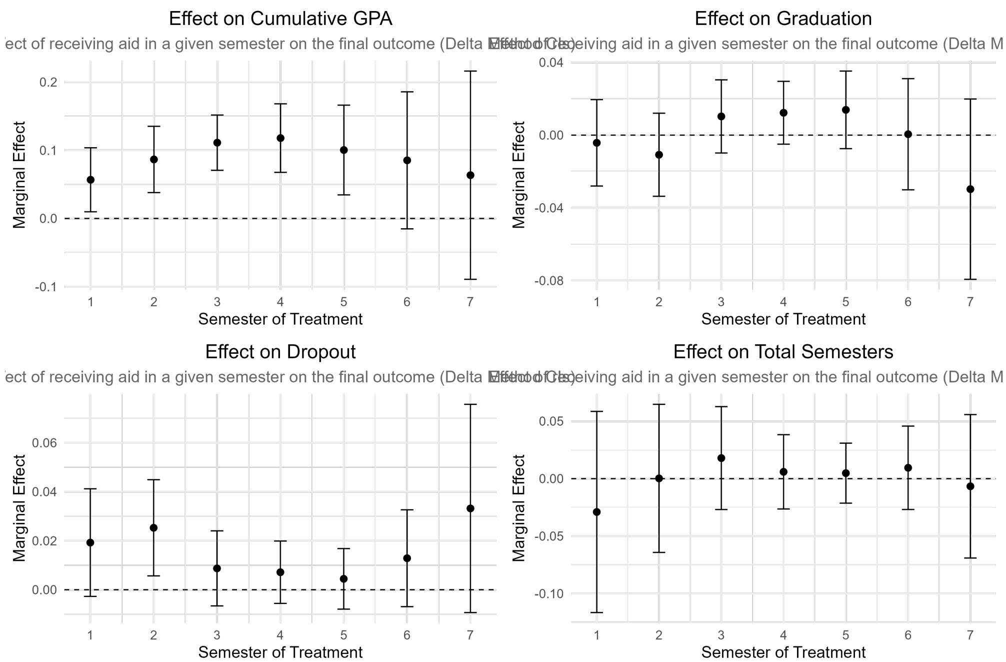
Notes: These figures show the results of the interaction term between past aid and receiving it in the fall for one-shot estimator for stipends for undergraduate students with a program length of 8 semesters. The figures report the estimates on stipend receipt in past on current grades, with the x axis reporting the lag.

Figure 22: One-shot estimator heterogeneity for stipends for those guaranteed aid



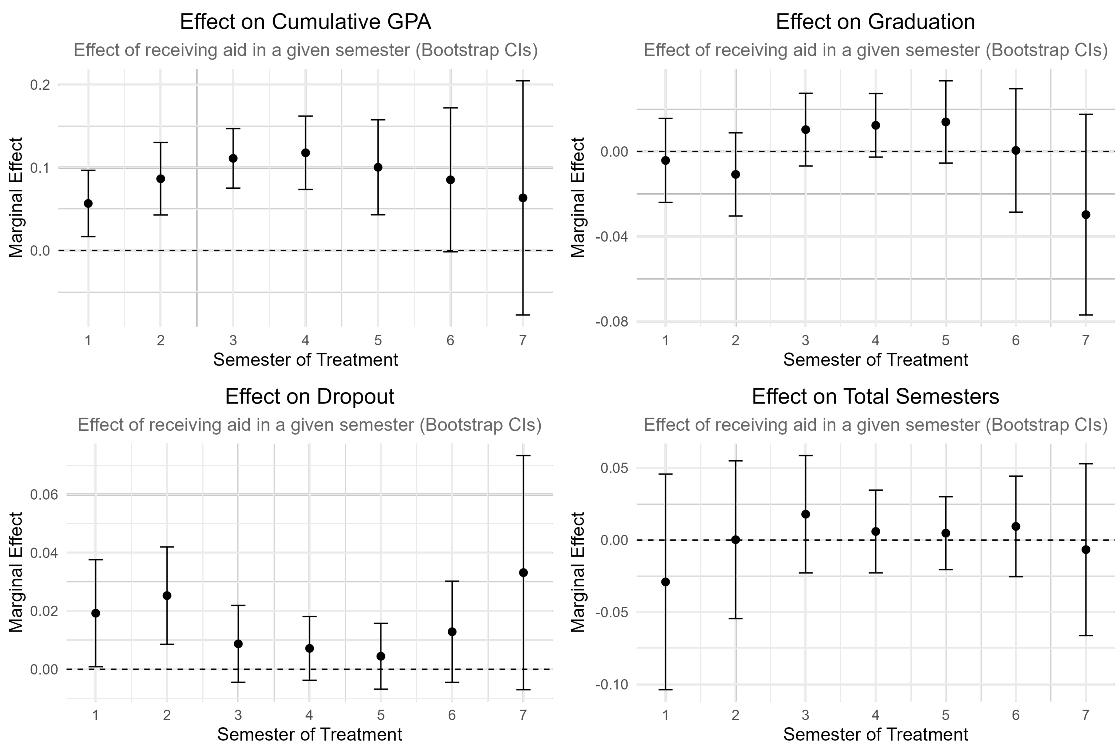
Notes: These figures show the results of the interaction term between past aid and having a guaranteed waiver for one-shot estimator for stipends for undergraduate students with a program length of 8 semesters. The figures report the estimates on stipend receipt in past on current grades, with the x axis reporting the lag.

Figure 23: Marginal Effect of Stipend Eligibility at a Given Semester with Delta Method SEs



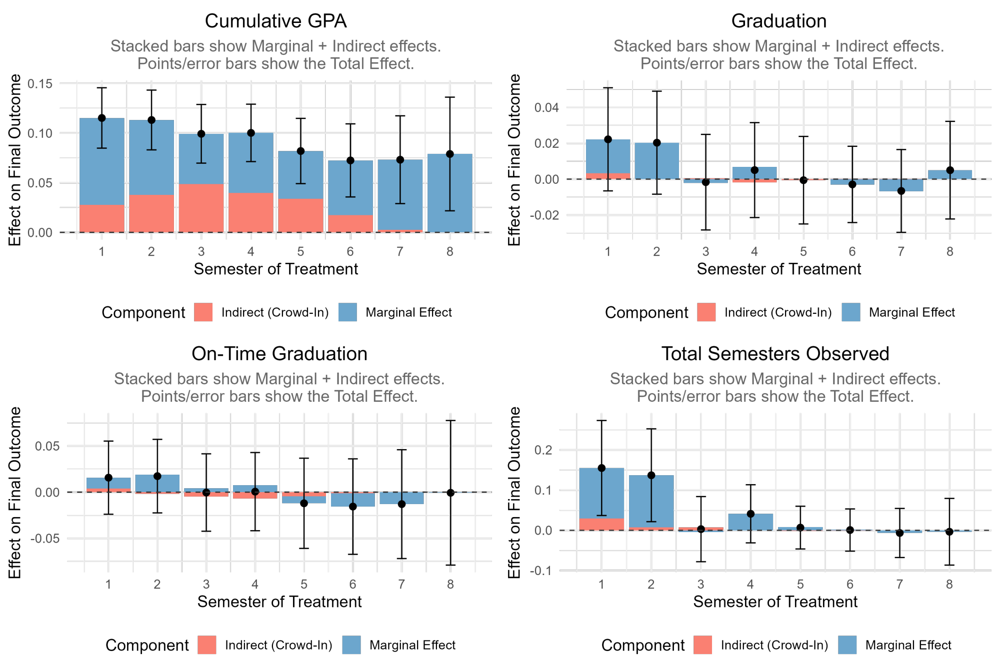
Notes: The figures report the estimates from an indirect regression discontinuity decomposition on the marginal impact of being eligible for stipends in any given semester on a range of student outcomes. The estimation is done on students who start during the time period and whose expected graduation falls before the data horizon, and whose program length is 8 semesters.

Figure 24: Marginal Effect of Stipend Eligibility at a Given Semester with Bootstrapped SEs



Notes: The figures report the estimates from an indirect regression discontinuity decomposition on the marginal, indirect, and total impact of being eligible for stipends in any given semester on a range of student outcomes. The estimation is done on students who start during the time period and whose expected graduation falls before the data horizon, and whose program length is 8 semesters.

Figure 25: Decomposition of Impact of Stipend Eligibility



Notes: The figures report the estimates from an indirect regression discontinuity decomposition on the marginal, indirect, and total impact of being eligible for stipends in any given semester on a range of student outcomes. The estimation is done on students who start during the time period and whose expected graduation falls before the data horizon, and whose program length is 8 semesters.

List of Tables

Table 1: Summary Statistics for RSU

	Full			Stipend			Waiver		
	Mean	Std.Dev.	N	Mean	Std.Dev.	N	Mean	Std.Dev.	N
GPA	0.00	0.98	103203	0.23	0.89	16360	−0.25	0.96	1864
Persisted	0.76	0.43	134991	0.98	0.15	16360	0.95	0.21	1864
On Track	0.79	0.41	134334	0.98	0.14	16216	0.96	0.21	1863
Dropout	0.23	0.42	134991	0.07	0.26	16360	0.12	0.32	1864
Graduation	0.66	0.47	95778	0.90	0.30	12643	0.80	0.40	1263
Graduation on Time	0.53	0.50	94998	0.80	0.40	12610	0.68	0.47	1263
Male	0.09	0.28	134991	0.12	0.32	16360	0.13	0.34	1864
Age	23.20	6.62	117243	21.79	5.81	15915	21.99	5.99	1827
Minority	0.11	0.31	117340	0.13	0.33	15919	0.15	0.35	1827
Capital	0.38	0.48	117340	0.37	0.48	15919	0.40	0.49	1827
Yearly Tuition fee	5540.25	483.61	82152	5492.41	634.83	15534	5380.15	695.33	1598
Tuition Waiver	0.38	0.49	134991	0.99	0.13	16360	0.43	0.49	1864
Stipend	0.05	0.22	134991	0.21	0.41	16360	0.04	0.19	1864
Cutoff Percentile			0	0.62	0.24	16360	0.40	0.26	1864

Note: The table reports the summary statistics for individual-semester dyads in the sample. The Tuition Waiver and Stipend marginal samples report the data for individuals one point away from the last person in their risk set who was awarded either a tuition waiver or stipend. Tuition fees are in Euros per year. The Cutoff Percentile is the percentile of the student in the risk set with the lowest GPA that received financial aid. The ontrack variable is an indicator of being on track in the next semester defined as the semester number that the student attends in half a year being larger by 1 than the current semester. The persistence variable is an indicator of completing the next semester. The minority variable is an indicator of previously attending a minority language of instruction school. The capital variable is an indicator of being from the capital city. The age variable is the age of the student upon enrolment.

Table 2: Summary Statistics for UL

	Full Sample			Marginal Sample		
	Mean	Std.Dev.	N	Mean	Std.Dev.	N
GPA	0.00	0.95	57856	0.36	0.85	24231
Persisted	0.61	0.49	63295	0.66	0.48	24231
On Track	0.39	0.49	63295	0.48	0.50	24231
Graduation	0.49	0.50	60634	0.59	0.49	23330
Graduation on Time	0.32	0.47	60634	0.41	0.49	23330
Stipend	0.31	0.46	63295	0.42	0.49	24231
Cutoff Percentile			0	0.74	0.18	24231

Note: The table reports summary statistics for individual-semester dyads. The Marginal Sample includes observations within the optimal bandwidth of the stipend eligibility cutoff. GPA is the raw, non-standardized grade. Stipend is an indicator for receiving a stipend in the current semester.

Table 3: Balance Test

	No FE	FE
Stipends		
Male	0.056***	-0.009
	(0.011)	(0.010)
Age	-2.111***	-0.081
	(0.208)	(0.170)
Riga	0.037*	0.018
	(0.016)	(0.015)
Minority	0.056***	0.023*
	(0.010)	(0.010)
Risk Set FE		X
Num.Obs.	16360	16360
Waivers		
Male	0.021	-0.017
	(0.032)	(0.031)
Age	-0.439	0.310
	(0.577)	(0.408)
Riga	-0.051	-0.063
	(0.047)	(0.047)
Minority	0.034	0.017
	(0.032)	(0.034)
Risk Set FE		X
Num.Obs.	1864	1864

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: Results from a regression of student characteristics on an indicator variable of being above or below the merit aid threshold and GPA in the previous semester. The columns report with and without inclusion of risk-set fixed effects. Being above the threshold is defined as having the at least as high a GPA as the student with the lowest GPA that got aid in a given calendar year, program, semester, and program year. Estimates done with cluster-robust standard errors clustered at the student level.

Table 4: Effect of waivers on Student Outcomes in the Next Semester

	OLS	Risk Set FE	2SLS	Per €1k
GPA in Next Semester	0.336***	0.378***	0.488***	0.160***
	(0.099)	(0.091)	(0.118)	(0.046)
Persisted in Next Semester	0.109***	0.102***	0.133***	0.056***
	(0.022)	(0.022)	(0.029)	(0.012)
On Track in Next Semester	0.094***	0.089***	0.115***	0.049***
	(0.021)	(0.022)	(0.028)	(0.012)
Graduation	0.139**	0.124**	0.154**	0.071***
	(0.049)	(0.041)	(0.051)	(0.020)
Dropped Out	−0.128***	−0.115***	−0.150***	−0.069***
	(0.035)	(0.032)	(0.041)	(0.017)
Graduation on Time	0.171**	0.116*	0.145*	0.072**
	(0.054)	(0.050)	(0.062)	(0.025)
Risk Set FE		X	X	X
Num.Obs.	1864	1864	1864	1598

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: The table reports estimates for various student outcomes. 'OLS' is the simple reduced-form estimate. 'Risk Set FE' includes fixed effects. '2SLS' instruments aid receipt with eligibility at the GPA cutoff. The average first-stage F-statistic across the 2SLS models is 525.2, indicating a strong instrument.

Table 5: Effect of Stipends on Student Outcomes in the Next Semester

	OLS	Risk Set FE	2SLS	Per €1k
GPA in Next Semester	0.175***	0.175***	0.306***	0.569***
	(0.028)	(0.028)	(0.048)	(0.090)
Persisted in Next Semester	0.010*	0.002	0.004	0.007
	(0.004)	(0.004)	(0.008)	(0.014)
On Track in Next Semester	0.013***	0.003	0.005	0.009
	(0.004)	(0.004)	(0.007)	(0.013)
Graduation	0.055***	−0.004	−0.006	−0.012
	(0.012)	(0.010)	(0.018)	(0.034)
Dropped Out	−0.024**	0.011	0.020	0.037
	(0.008)	(0.008)	(0.014)	(0.025)
Graduation on Time	0.065***	0.024+	0.042+	0.081+
	(0.015)	(0.013)	(0.024)	(0.046)
Risk Set FE		X	X	X
Num.Obs.	16 360	16 360	16 360	16 360

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: The table reports estimates for various student outcomes. 'OLS' is the simple reduced-form estimate. 'Risk Set FE' includes fixed effects. '2SLS' instruments aid receipt with eligibility at the GPA cutoff. The average first-stage F-statistic across the 2SLS models is 2992.2, indicating a strong instrument.

Table 6: Effect of Stipends on Student Outcomes in the Next Semester in UL

	OLS	Risk Set FE	2SLS	Per €1k
GPA in Next Semester	0.253***	0.204***	0.213***	0.428***
	(0.026)	(0.025)	(0.026)	(0.053)
Persisted in Next Semester	0.066***	0.066***	0.069***	0.135***
	(0.011)	(0.008)	(0.009)	(0.017)
On Track in Next Semester	0.083***	0.076***	0.080***	0.182***
	(0.012)	(0.011)	(0.012)	(0.021)
Graduation	0.116***	0.088***	0.093***	0.156***
	(0.012)	(0.010)	(0.011)	(0.023)
Graduation on Time	0.091***	0.070***	0.074***	0.145***
	(0.012)	(0.011)	(0.012)	(0.023)
Risk Set FE		X	X	X
Num.Obs.	24 231	24 231	24 231	24 231

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: The table reports estimates for various student outcomes. 'OLS' is the simple reduced-form estimate. 'Risk Set FE' includes fixed effects. '2SLS' instruments aid receipt with eligibility at the GPA cutoff. The average first-stage F-statistic across the 2SLS models is 61032.3, indicating a strong instrument.

Table 7: Heterogeneity by Student Demographics in the Effects of Aid on Student Outcomes

	Heterogeneity Variable			
Outcome	Male	Age	Capital	Minority
Stipends				
GPA	0.167+	-0.003	-0.111*	0.006
	(0.093)	(0.005)	(0.057)	(0.080)
Persistence	-0.019	0.001	0.019*	0.008
	(0.014)	(0.001)	(0.009)	(0.012)
On Track	-0.007	0.001	0.017*	0.001
	(0.013)	(0.001)	(0.009)	(0.011)
Dropout	0.033	-0.001	-0.016	-0.027
	(0.028)	(0.001)	(0.016)	(0.025)
Graduation	-0.056	0.003	0.009	0.027
	(0.036)	(0.002)	(0.020)	(0.030)
On-time Grad.	0.031	0.002	0.016	0.077+
	(0.056)	(0.003)	(0.027)	(0.041)
Aid Receipt	-0.119***	0.001	-0.021	-0.010
	(0.026)	(0.002)	(0.017)	(0.025)
Risk Set FE	X	X	X	X
Num.Obs.	16360	15915	15919	15919
Waivers				
GPA	-0.013	0.008	0.016	-0.277
	(0.291)	(0.015)	(0.184)	(0.283)
Persistence	-0.076	0.005	0.001	0.064
	(0.061)	(0.004)	(0.047)	(0.072)
On Track	-0.066	0.005	0.004	0.062
	(0.061)	(0.004)	(0.045)	(0.072)
Dropout	0.108	0.002	0.050	-0.040
	(0.083)	(0.005)	(0.064)	(0.088)
Graduation	0.003	-0.004	-0.015	0.034
	(0.108)	(0.006)	(0.080)	(0.097)
On-time Grad.	0.024	-0.007	0.024	-0.004
	(0.128)	(0.006)	(0.093)	(0.147)
Aid Receipt	-0.155*	-0.003	0.103*	0.058
	(0.074)	(0.004)	(0.049)	(0.070)
Risk Set FE	X	X	X	X
Num.Obs.	1864	1827	1827	1827

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: The table reports the estimates from a series of regressions estimating the impact of being eligible for financial aid on student outcomes. Each estimate is from a separate regression. Persistence is defined as the student still being enrolled in a given semester. Robust standard errors clustered at the student level in parentheses.

Table 8: Heterogeneity by Cutoff Percentile in the Effects of Aid on Student Outcomes

	Heterogeneity Variable			
Outcome	Cutoff Percentile	First Half	Last Year	Last Semester
Stipends				
GPA	-0.356**	0.005	0.131*	-0.095
	(0.135)	(0.055)	(0.054)	(0.081)
Persistence	-0.022	0.005	-0.011	0.013
	(0.025)	(0.008)	(0.008)	(0.012)
On Track	-0.011	0.012	-0.013+	0.004
	(0.023)	(0.008)	(0.007)	(0.012)
Dropout	0.062+	0.011	-0.017	-0.010
	(0.035)	(0.013)	(0.014)	(0.016)
Graduation	-0.079+	-0.014	0.000	0.016
	(0.047)	(0.018)	(0.018)	(0.019)
On-time Grad.	-0.163**	-0.031	-0.000	0.019
	(0.062)	(0.025)	(0.024)	(0.030)
Aid Receipt	0.670***	0.050**	0.014	-0.002
	(0.036)	(0.018)	(0.017)	(0.024)
Risk Set FE	X	X	X	X
Num.Obs.	16360	16327	16327	16327
Waivers				
GPA	-0.498	0.091	0.146	0.272
	(0.378)	(0.190)	(0.182)	(0.304)
Persistence	-0.308**	0.075+	-0.152***	0.004
	(0.095)	(0.040)	(0.040)	(0.059)
On Track	-0.312***	0.066+	-0.129***	-0.004
	(0.091)	(0.040)	(0.039)	(0.063)
Dropout	0.109	-0.127*	0.064	0.139*
	(0.131)	(0.057)	(0.063)	(0.059)
Graduation	-0.230	0.199*	0.065	-0.304***
	(0.181)	(0.079)	(0.082)	(0.086)
On-time Grad.	-0.275	0.048	0.043	-0.142
	(0.218)	(0.097)	(0.112)	(0.133)
Aid Receipt	0.179*	-0.017	0.098*	-0.196*
	(0.085)	(0.048)	(0.043)	(0.080)
Risk Set FE	X	X	X	X
Num.Obs.	1864	1864	1864	1864

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: The table reports the estimates from a series of regressions estimating the impact of being eligible financial aid on student grades, persistence, being on track in the next semester, dropout, graduation and graduation on time. Each estimate is from a separate regression. Persistence is defined as the student still being enrolled in a given semester. Robust standard errors clustered at the student level in parentheses.

Table 9: Heterogeneity by Cutoff Chracteristics for UL

	Heterogeneity Variable			
	Cutoff Percentile	First Half	Last Year	Last Semester
GPA	-0.064	-0.187^{***}	0.083	0.791
	(0.150)	(0.050)	(0.054)	(0.586)
Persistence	-0.016	0.045^{**}	-0.042^{**}	-0.071^{***}
	(0.054)	(0.016)	(0.015)	(0.011)
On Track	-0.052	0.028	-0.017	-0.090^{**}
	(0.062)	(0.022)	(0.023)	(0.031)
Graduation	-0.044	0.016	-0.013	-0.085^{***}
	(0.059)	(0.020)	(0.019)	(0.017)
Ontime	-0.026	0.030	-0.019	-0.083^{**}
	(0.057)	(0.021)	(0.022)	(0.030)
Aid Receipt	0.400^{***}	0.000	0.010	0.002
	(0.035)	(0.007)	(0.007)	(0.010)
Risk Set FE	X	X	X	X
Num.Obs.	24 231	24 231	24 231	24 231

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: The table reports the estimates from a series of regressions estimating the impact of being eligible financial aid on student grades, persistence, being on track in the next semester, graduation and graduation on time. Each estimate is from a separate regression. Persistence is defined as the student still being enrolled in a given semester. Robust standard errors clustered at the student level in parentheses.

Table 10: Heterogeneity by Competitiveness in the Effects of Aid on Student Outcomes in RSU

	Heterogeneity Variable			
Outcome	Fall	<Median GPA	<Median Progression	Guaranteed Aid
Stipends				
GPA	-0.030	0.128*	-0.047	0.198+
	(0.053)	(0.055)	(0.061)	(0.109)
Persistence	-0.012	-0.015+	0.016	-0.007
	(0.008)	(0.008)	(0.015)	(0.015)
On Track	-0.012	-0.021**	0.011	-0.009
	(0.008)	(0.008)	(0.014)	(0.015)
Dropout	-0.003	0.007	0.008	-0.015
	(0.014)	(0.015)	(0.019)	(0.029)
Graduation	0.019	-0.031	-0.008	0.008
	(0.017)	(0.020)	(0.025)	(0.045)
Ontime	0.027	-0.048+	-0.007	-0.009
	(0.024)	(0.027)	(0.032)	(0.058)
Risk Set FE	X	X	X	X
Num.Obs.	16360	16360	16360	16360
Waivers				
GPA	0.145	0.350+	0.493**	
	(0.196)	(0.184)	(0.183)	
Persistence	-0.090+	0.089*	0.260***	
	(0.051)	(0.044)	(0.055)	
On Track	-0.063	0.076+	0.246***	
	(0.048)	(0.043)	(0.055)	
Dropout	0.053	-0.049	-0.147*	
	(0.072)	(0.057)	(0.065)	
Graduation	-0.069	0.049	0.098	
	(0.082)	(0.070)	(0.089)	
Ontime	-0.006	-0.037	0.143	
	(0.107)	(0.088)	(0.110)	
Risk Set FE	X	X	X	
Num.Obs.	1864	1864	1864	

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: The table reports the estimates from a series of regressions estimating the impact of being eligible for financial aid on student grades, persistence, being on track in the next semester, dropout, graduation and graduation on time. Each estimate is from a separate regression. High difficulty is defined as a risk set being below the median GPA or rate of students progressing on track. Persistence is defined as the student still being enrolled in a given semester. Guaranteed aid is for individuals who started their studies with a tuition waiver and thus have it guaranteed for the rest of their studies. Robust standard errors clustered at the student level in parentheses.

Table 11: Heterogeneity by Competitiveness in the Effects of Stipends on Student Outcomes in UL

	Heterogeneity Variable		
	Fall	<Median GPA	<Median Progression
GPA	−0.057	−0.141**	0.053
	(0.055)	(0.050)	(0.051)
Persistence	0.026	0.019	0.032+
	(0.017)	(0.018)	(0.017)
On Track	0.060**	−0.016	−0.005
	(0.021)	(0.023)	(0.022)
Graduation	0.041*	−0.036+	0.007
	(0.019)	(0.021)	(0.020)
Ontime	0.051*	−0.016	−0.012
	(0.021)	(0.022)	(0.021)
Aid Receipt	0.023***	−0.017*	−0.002
	(0.007)	(0.007)	(0.007)
Risk Set FE	X	X	X
Num.Obs.	24 231	24 231	24 231

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

[l]Note: The table reports the estimates from a series of regressions estimating
the impact of being eligible for financial aid on student outcomes, interacted
with various measures of competition. Each estimate is from a separate regression.
High difficulty is defined as a risk set being below the median raw GPA or,
rate of students progressing on track. Robust standard errors clustered at the student level in parentheses.

Table 12: Waiver Effects In Future Semesters

	Dependent Variable				
Outcome	Waivers	GPA	Persistence	On Track	Aid in 000s of EUR
Next Semester	0.769***	0.378***	0.102***	0.089***	2.109***
	(0.023)	(0.091)	(0.022)	(0.022)	(0.067)
2 semesters ahead	0.434***	0.344**	0.106***	0.147***	1.335***
	(0.033)	(0.109)	(0.025)	(0.031)	(0.099)
3 semesters ahead	0.113**	0.048	0.056*	0.151***	0.340**
	(0.038)	(0.148)	(0.027)	(0.032)	(0.117)
4 semesters ahead	0.125*	0.296+	0.085***	0.170***	0.324*
	(0.052)	(0.158)	(0.026)	(0.033)	(0.140)
5 semesters ahead	0.146**	0.032	0.062**	0.124***	0.388**
	(0.056)	(0.233)	(0.022)	(0.030)	(0.150)
6 semesters ahead	0.044	−0.059	0.014	0.114***	0.133
	(0.078)	(0.224)	(0.022)	(0.030)	(0.209)
Risk Set FE	X	X	X	X	X
Num.Obs.	1864	1661	1864	1863	1598

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: The table reports the estimates from a series of regressions estimating the impact of being offered waivers on student grades, persistence, and reception of aid again in future semesters. Each estimate is from a separate regression. Persistence is defined as the student still being enrolled in a given semester. Robust standard errors clustered at the student level in parentheses.

Table 13: Stipend Effects In Future Semesters for RSU

	Dependent Variable				
Outcome	Stipends	GPA	Persistence	On Track	Aid in 000s of EUR
Next Semester	0.557***	0.175***	0.002	0.003	0.302***
	(0.010)	(0.028)	(0.004)	(0.004)	(0.006)
2 semesters ahead	0.119***	0.176***	0.001	0.009	0.073***
	(0.011)	(0.033)	(0.006)	(0.006)	(0.006)
3 semesters ahead	0.064***	0.096*	−0.017**	0.001	0.037***
	(0.012)	(0.038)	(0.006)	(0.007)	(0.007)
4 semesters ahead	0.053***	0.070	−0.012*	0.001	0.029***
	(0.013)	(0.048)	(0.006)	(0.008)	(0.008)
5 semesters ahead	0.051**	0.086	0.004	0.006	0.031**
	(0.016)	(0.054)	(0.005)	(0.008)	(0.010)
6 semesters ahead	0.032+	0.147*	−0.003	0.004	0.014
	(0.019)	(0.068)	(0.005)	(0.008)	(0.011)
Risk Set FE	X	X	X	X	X
Num.Obs.	16 360	14 935	16 360	16 216	16 360

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: The table reports the estimates from a series of regressions estimating the impact of being offered stipends on student grades, persistence, and reception of aid again in future semesters. Each estimate is from a separate regression. Persistence is defined as the student still being enrolled in a given semester. Robust standard errors clustered at the student level in parentheses.

Table 14: Stipend Effects In Future Semesters for UL

	Dependent Variable				
	Stipends	GPA	Persistence	On Track	Aid in 000s of EUR
Next Semester	0.949***	0.204***	0.066***	0.076***	0.489***
	(0.004)	(0.025)	(0.008)	(0.011)	(0.002)
2 semesters ahead	0.116***	0.206***	0.063***	0.071***	0.059***
	(0.013)	(0.033)	(0.009)	(0.011)	(0.007)
3 semesters ahead	0.140***	0.198***	0.047***	0.072***	0.073***
	(0.018)	(0.050)	(0.007)	(0.011)	(0.010)
4 semesters ahead	0.092***	0.204**	0.023***	0.071***	0.045**
	(0.027)	(0.075)	(0.006)	(0.011)	(0.015)
5 semesters ahead	0.097*	0.101	0.011**	0.067***	0.058*
	(0.042)	(0.124)	(0.004)	(0.011)	(0.023)
6 semesters ahead	0.046	−0.019	0.005+	0.067***	0.026
	(0.067)	(0.219)	(0.003)	(0.011)	(0.035)
Risk Set FE	X	X	X	X	X
Num.Obs.	24 231	15 834	24 231	24 231	24 231

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: The table reports the estimates from a series of regressions estimating the impact of being offered stipends on student grades, persistence, and reception of aid again in future semesters. Each estimate is from a separate regression. Persistence is defined as the student still being enrolled in a given semester. Robust standard errors clustered at the student level in parentheses.

Table 15: Dynamic Complementarity in Skills Development

	Waivers	Stipends
Current CGPA	0.751***	0.829***
	(0.020)	(0.006)
Aid	0.104***	0.017***
	(0.019)	(0.005)
Aid * Current CGPA	0.124***	0.043***
	(0.022)	(0.007)
Risk Set FE	X	X
Num.Obs.	3268	34 191

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: Results from a two stage least squares regression of the next period CGPA on the current period CGPA and an interaction term of aid and current period CGPA. The regression uses risk set fixed effects. Being above the threshold is defined as having the at least as high a GPA as the student with the lowest GPA that got aid in a given calendar year, program, semester, and program year. Estimates done with robust standard errors clustered at the student level.

Table 16: Dynamic Complementarity in Skills Development for UL

	(1)
Current CGPA	0.807***
	(0.006)
Aid	0.082***
	(0.010)
Aid * Current CGPA	0.019+
	(0.011)
Risk Set FE	X
Num.Obs.	23 645

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: Results from a two stage least squares regression of the next period CGPA on the current period CGPA and an interaction term of aid and current period CGPA. The regression uses risk set fixed effects. Being above the threshold is defined as having the at least as high a GPA as the student with the lowest GPA that got aid in a given calendar year, program, semester, and program year. Estimates done with robust standard errors clustered at the student level.

Table 17: Estimation of Stipend Eligibility Effects with Different Bandwidths for RSU

	Bandwidth				
	(1)	(2)	(3)	(4)	(5)
GPA	0.132***	0.179***	0.179***	0.229***	0.326***
	(0.039)	(0.028)	(0.028)	(0.026)	(0.023)
Persistence	0.002	0.002	0.002	0.000	0.000
	(0.006)	(0.004)	(0.004)	(0.004)	(0.003)
On Track	−0.001	0.003	0.003	0.000	0.001
	(0.006)	(0.004)	(0.004)	(0.003)	(0.003)
Graduation	0.004	−0.004	−0.004	−0.005	0.010
	(0.014)	(0.010)	(0.010)	(0.009)	(0.008)
Dropout	−0.002	0.011	0.011	0.008	0.000
	(0.011)	(0.008)	(0.008)	(0.007)	(0.006)
Graduation on Time	0.046*	0.025+	0.025+	0.007	0.021+
	(0.018)	(0.013)	(0.013)	(0.013)	(0.012)
Risk Set FE	9967.000	15 352.000	15 352.000	18 412.000	21 868.000

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: The table reports the estimates from regressions from a reduced-form regression for various bandwidths. Robust standard errors clustered at the student level in parentheses.

Table 18: Estimation of Waiver Eligibility Effects with Different Bandwidths

	Bandwidth				
	(1)	(2)	(3)	(4)	(5)
GPA	0.326+	0.357+	0.420***	0.378***	0.418***
	(0.193)	(0.208)	(0.091)	(0.101)	(0.075)
Persistence	0.107**	0.095+	0.107***	0.079**	0.091***
	(0.038)	(0.049)	(0.022)	(0.025)	(0.017)
On Track	0.070+	0.072	0.094***	0.065**	0.081***
	(0.039)	(0.046)	(0.022)	(0.024)	(0.017)
Graduation	0.221*	0.267**	0.132**	0.148***	0.087*
	(0.085)	(0.088)	(0.042)	(0.043)	(0.036)
Dropout	−0.188**	−0.167*	−0.130***	−0.131***	−0.090***
	(0.060)	(0.072)	(0.033)	(0.035)	(0.027)
Graduation on Time	0.148	0.235*	0.122*	0.113*	0.102*
	(0.107)	(0.110)	(0.050)	(0.057)	(0.041)
Risk Set FE	458.000	771.000	1592.000	1418.000	1952.000

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: The table reports the estimates from regressions from a reduced-form regression for various bandwidths. Robust standard errors clustered at the student level in parentheses.

Table 19: Estimation of First Stage with Various Cutoffs for RSU

	Cutoff Value				
	-2	-1	0	1	2
Stipends					
Stipend	-0.076***	0.028+	0.557***	0.449***	0.304***
	(0.016)	(0.016)	(0.010)	(0.010)	(0.010)
Num.Obs.	15401	15978	16360	16705	17075
Waivers					
Waiver	-0.325***	-0.140***	0.769***	0.508***	0.243***
	(0.041)	(0.039)	(0.023)	(0.030)	(0.028)
Num.Obs.	1776	1826	1864	1854	1778
Risk Set FE	X	X	X	X	X

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: The table reports the estimates from regressions of student of receiving aid on whether they are above the cutoff or not for various cutoff values with risk set fixed effects. Robust standard errors clustered at the student level in parentheses.

Table 20: Estimation of Stipend Eligibility Effects with Different Bandwidths for UL

	Bandwidth				
	(1)	(2)	(3)	(4)	(5)
GPA	0.187***	0.204***	0.285***	0.364***	0.410***
	(0.027)	(0.025)	(0.021)	(0.019)	(0.018)
Persistence	0.064***	0.066***	0.061***	0.065***	0.073***
	(0.009)	(0.008)	(0.007)	(0.006)	(0.006)
On Track	0.071***	0.076***	0.093***	0.103***	0.119***
	(0.012)	(0.011)	(0.009)	(0.009)	(0.008)
Graduation	0.086***	0.088***	0.087***	0.093***	0.103***
	(0.011)	(0.010)	(0.009)	(0.008)	(0.007)
Graduation on Time	0.064***	0.070***	0.092***	0.103***	0.119***
	(0.012)	(0.011)	(0.009)	(0.009)	(0.008)
Risk Set FE	21 907.000	24 220.000	30 922.000	35 760.000	38 629.000

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: The table reports the estimates from regressions from a reduced-form regression for various bandwidths. Robust standard errors clustered at the student level in parentheses.

Table 21: Estimation of First Stage with Various Cutoffs for UL

	Cutoff Value				
	-2	-1	0	1	2
Stipend	-0.250^{***}	0.157^{***}	0.949^{***}	0.541^{***}	0.182^{***}
	(0.007)	(0.008)	(0.004)	(0.007)	(0.006)
Risk Set FE	X	X	X	X	X
Num.Obs.	21 201	22 828	24 231	25 326	26 125

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: The table reports the estimates from regressions of student of receiving aid on whether they are above the cutoff or not for various cutoff values with risk set fixed effects. Robust standard errors clustered at the student level in parentheses.

Table 22: Balance Test using RDHonest

	Male	Age	Capital	Minority
Stipends				
RD Estimate (Honest)	-0.022*	-0.233	0.026	0.010
	(0.011)	(0.216)	(0.021)	(0.011)
Waivers				
RD Estimate (Honest)	-0.075	0.223	0.032	-0.067
	(0.081)	(0.474)	(0.074)	(0.059)

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 23: Effect of Aid on Student Outcomes using Armstrong and Kolesar (2020) CIs

	GPA	Persistence	On Track	Graduation	Dropout	On Time
Stipends						
RD Estimate	0.134**	0.002	0.004	-0.005	0.012	0.027+
	(0.036)	(0.004)	(0.003)	(0.010)	(0.007)	(0.014)
Waivers						
RD Estimate	0.297*	0.083**	0.063*	0.132**	-0.174*	0.104
	(0.128)	(0.027)	(0.026)	(0.044)	(0.059)	(0.054)

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: The table reports, bias-corrected RD estimates with optimal bandwidths selected using the MSE-optimal criterion in accordance with the procedure suggested by Armstrong and Kolesár (2020). Each estimate is from a separate regression of an outcome on the running variable, with risk-set fixed effects.

Table 24: Effect of Aid on Student Outcomes in LU

	(1)
GPA	
RD Estimate	0.129***
	(0.033)
N (Effective)	10847
Persistence	
RD Estimate	0.061***
	(0.010)
N (Effective)	19278
On Track	
RD Estimate	0.071***
	(0.011)
N (Effective)	21939
Graduation	
RD Estimate	0.086***
	(0.010)
N (Effective)	21145
Graduation on Time	
RD Estimate	0.064***
	(0.011)
N (Effective)	21145

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Results from a regression of student outcomes on an indicator variable of being above or below the merit aid threshold and GPA in the previous semester with a bandwidth of 6 ranks. The regression uses the RD procedure suggested by Armstrong and Kolesár (2020) Each regression includes risk set fixed effects. Being above the threshold is defined as having the at least as high a GPA as the student with the lowest GPA that got aid in a given calendar year, program, semester, and program year. Estimates done with cluster-robust standard errors clustered at the student level.

Table 25: Effect of Aid on Student Outcomes in LU (IV)

	(1)
GPA	
RD Estimate	0.133***
	(0.034)
N (Effective)	10847
Persistence	
RD Estimate	0.064***
	(0.010)
N (Effective)	19278
On Track	
RD Estimate	0.075***
	(0.012)
N (Effective)	21939
Graduation	
RD Estimate	0.090***
	(0.011)
N (Effective)	21145
Graduation on Time	
RD Estimate	0.068***
	(0.011)
N (Effective)	21145

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Results from a two stage least squares regression of the outcome variable on receiving aid. The regression uses the RD procedure suggested by Armstrong and Kolesár (2020) Each regression includes risk set fixed effects. Being above the threshold is defined as having the at least as high a GPA as the student with the lowest GPA that got aid in a given calendar year, program, semester, and program year. Estimates done with cluster-robust standard errors clustered at the student level.